EI331 Signals and Systems

Lecture 9

Bo Jiang

John Hopcroft Center for Computer Science
Shanghai Jiao Tong University

March 26, 2019

Contents

1. Eigenvalues and Eigenfunctions

2. CT Fourier Series

3. Properties of CT Fourier Series

Recap

	$\mathbb{R}^n$	DT signals $\mathbb{C}^{\mathbb{Z}}$	CT signals
basis	$\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$	$\{\delta_k : k \in \mathbb{Z}\}$	$\{\delta_\tau : \tau \in \mathbb{R}\}$[1]
general x	$x = \sum_{k=1}^{n} x_k \mathbf{e}_k$	$x = \sum_{k \in \mathbb{Z}} x[k]\delta_k$	$x = \int_{\mathbb{R}} x(\tau)\delta_\tau d\tau$

Image under linear transformation

basis	$\{\mathbf{f}_1, \ldots, \mathbf{f}_n\}$	$\{h_k : k \in \mathbb{Z}\}$	$\{h_\tau : \tau \in \mathbb{R}\}$
general x	$Ax = \sum_{k=1}^{n} x_k \mathbf{f}_k$	$T(x) = \sum_{k \in \mathbb{Z}} x[k]h_k$	$T(x) = \int_{\mathbb{R}} x(\tau)h_\tau d\tau$
LTI	–	$T(x) = x * h$	$T(x) = x * h$

[1] Not standard usage of term "basis"; understood in terms of integral representation on next line.

Eigenvalues and Eigenvectors

In linear algebra, $v \in \mathbb{R}^n$ is eigenvector of matrix A with associated eigenvalue λ, if

$$Av = \lambda v$$

Suppose A has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ associated to eigenvectors $v_1, \ldots, v_n$, respectively.

- $v_1, \ldots, v_n$ form a basis of $\mathbb{R}^n$, i.e. $\forall x \in \mathbb{R}^n$ has following representation

$$x = \sum_{k=1}^{n} a_k v_k$$

- image under A

$$Ax = \sum_{k=1}^{n} a_k A v_k = \sum_{k=1}^{n} a_k \lambda_k v_k$$

Change of Basis

	standard basis	eigenbasis of A
basis vectors	$\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$	$\{v_1, \ldots, v_n\}$
decomposition	$x = \sum_{k=1}^{n} x_k \mathbf{e}_k$	$x = \sum_{k=1}^{n} a_k v_k$
image under A	$Ax = \sum_{k=1}^{n} x_k \mathbf{f}_k$	$Ax = \sum_{k=1}^{n} a_k \lambda_k v_k$

Expansions of x and Ax

- in standard basis: same coefficients $\{x_k\}$, different basis vectors $\{\mathbf{e}_k\} \mapsto \{\mathbf{f}_k\}$
- in eigenbasis: same basis vectors $\{v_k\}$, different coefficients $\{a_k\} \mapsto \{\lambda_k a_k\}$

Change of Basis

		standard basis	eigenbasis
DT	basis vectors	$\{\delta_k : k \in \mathbb{Z}\}$	?
	decomposition	$x = \sum_{k \in \mathbb{Z}} x[k]\delta_k$	
	image under T	$T(x) = \sum_{k \in \mathbb{Z}} x[k]h_k$	

		"standard basis"	eigenbasis
CT	basis vectors	$\{\delta_\tau : \tau \in \mathbb{R}\}$	?
	decomposition	$x = \int_{\mathbb{R}} x(\tau)\delta_\tau d\tau$	
	image under T	$T(x) = \int_{\mathbb{R}} x(\tau)h_\tau d\tau$	

Eigenvalues and Eigenfunctions of LTI Systems

CT Exponential e^{st}

$$T(e^{st}) = \int_{\mathbb{R}} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{\mathbb{R}} h(\tau) e^{-s\tau} d\tau,$$

e^{st} is eigenfunction of T with associated eigenvalue

$$H(s) = \int_{\mathbb{R}} h(\tau) e^{-s\tau} d\tau \quad \text{(system function)}$$

DT Exponential z^n

$$T(z^n) = \sum_{k \in \mathbb{Z}} h[k] z^{n-k} d\tau = z^n \sum_{k \in \mathbb{Z}} h[k] z^{-k},$$

z^n is eigenfunction of T with associated eigenvalue

$$H(z) = \sum_{k \in \mathbb{Z}} h[k] z^{-k} \quad \text{(system function)}$$

Response to Linear Combination of Eigenfunctions

CT
$$T\left(\sum_k a_k e^{s_k t}\right) = \sum_k a_k H(s_k) e^{s_k t}$$

DT
$$T\left(\sum_k a_k z_k^n\right) = \sum_k a_k H(z_k) z_k^n$$

Input vs. output

- linear combinations of same exponentials
- different coefficients $\{a_k\} \mapsto \{H(s_k)a_k\}/\{H(z_k)a_k\}$

Questions

- Which functions are linear combinations of exponentials?
- How to find coefficients a_k?

Fourier analysis focuses on $e^{j\omega t}$ for CT and $e^{j\omega n}$ for DT

Contents

1. Eigenvalues and Eigenfunctions

2. CT Fourier Series

3. Properties of CT Fourier Series

CT Fourier Series

Recall CT signal is periodic with period T if

$$x = \tau_T x \quad \text{or} \quad x(t) = x(t+T), \forall t \in \mathbb{R}$$

- fundamental period T: smallest positive period
- fundamental frequency $\omega_0 = \frac{2\pi}{T}$

$\sin(\omega_0 t)$, $\cos(\omega_0 t)$, $e^{j\omega_0 t}$ periodic with fundamental frequency ω_0

Fourier series represent periodic signals in terms of harmonically related sinusoids or complex exponentials

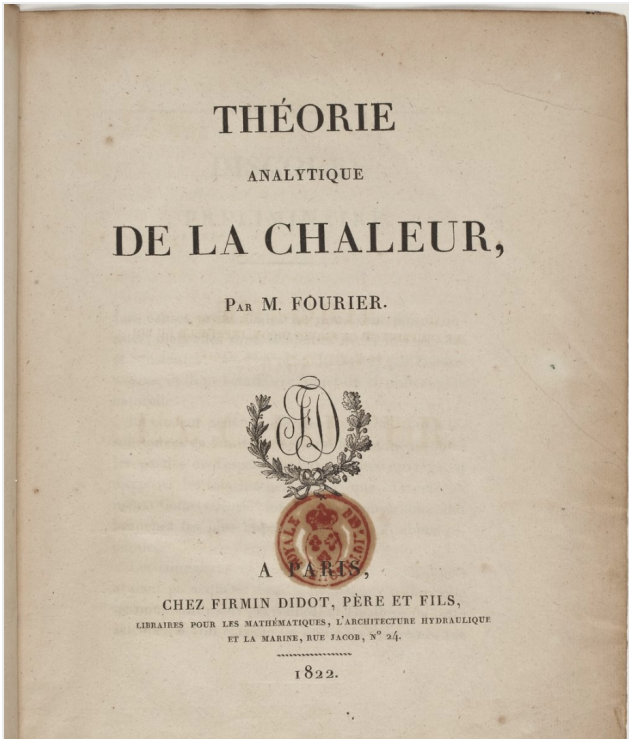
$$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)]$$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Joseph Fourier

1807, *Mémoire sur la propagation de la chaleur dans les corps solides* (*Memoir on the propagation of heat in solid bodies*)

1822, *Théorie analytique de la chaleur* (*The Analytical Theory of Heat*)



THÉORIE

ANALYTIQUE

DE LA CHALEUR,

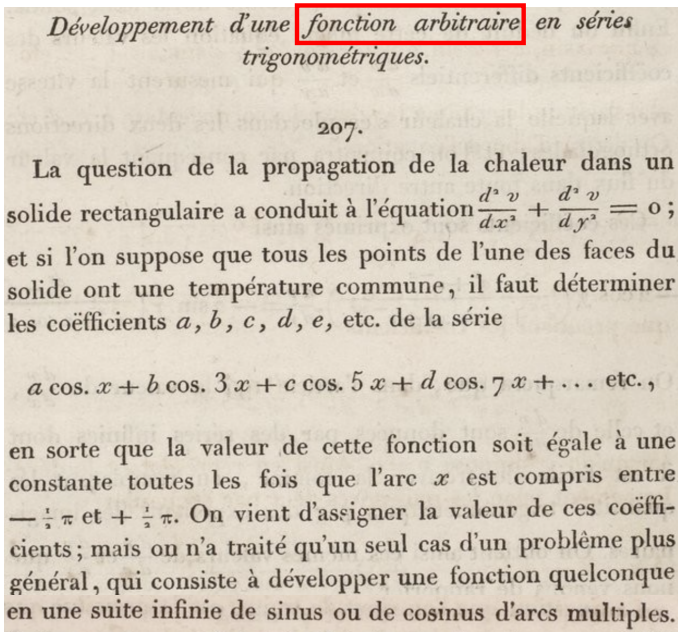
PAR M. FOURIER.

A PARIS,

CHEZ FIRMIN DIDOT, PÈRE ET FILS,

LIBRAIRES POUR LES MATHÉMATIQUES, L'ARCHITECTURE HYDRAULIQUE ET LA MARINE, RUE JACOB, N° 24.

1822.



Développement d'une fonction arbitraire en séries trigonométriques.

207.

La question de la propagation de la chaleur dans un solide rectangulaire a conduit à l'équation $\frac{d^2 v}{dx^2} + \frac{d^2 v}{dy^2} = 0$; et si l'on suppose que tous les points de l'une des faces du solide ont une température commune, il faut déterminer les coëfficients $a, b, c, d, e,$ etc. de la série

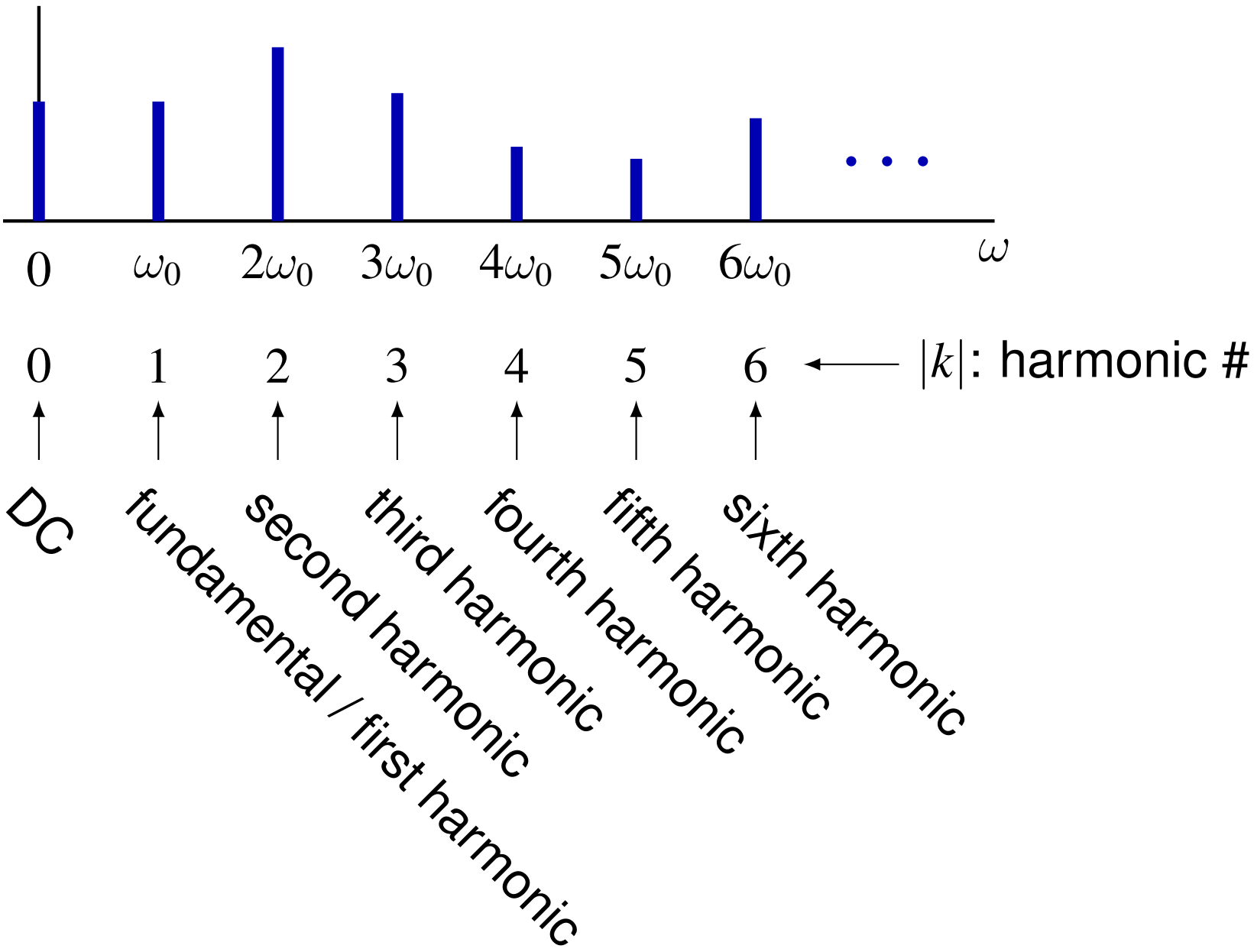
$$a \cos. x + b \cos. 3x + c \cos. 5x + d \cos. 7x + \ldots \text{ etc.},$$

en sorte que la valeur de cette fonction soit égale à une constante toutes les fois que l'arc x est compris entre $-\frac{1}{2}\pi$ et $+\frac{1}{2}\pi$. On vient d'assigner la valeur de ces coëfficients; mais on n'a traité qu'un seul cas d'un problême plus général, qui consiste à développer une fonction quelconque en une suite infinie de sinus ou de cosinus d'arcs multiples.

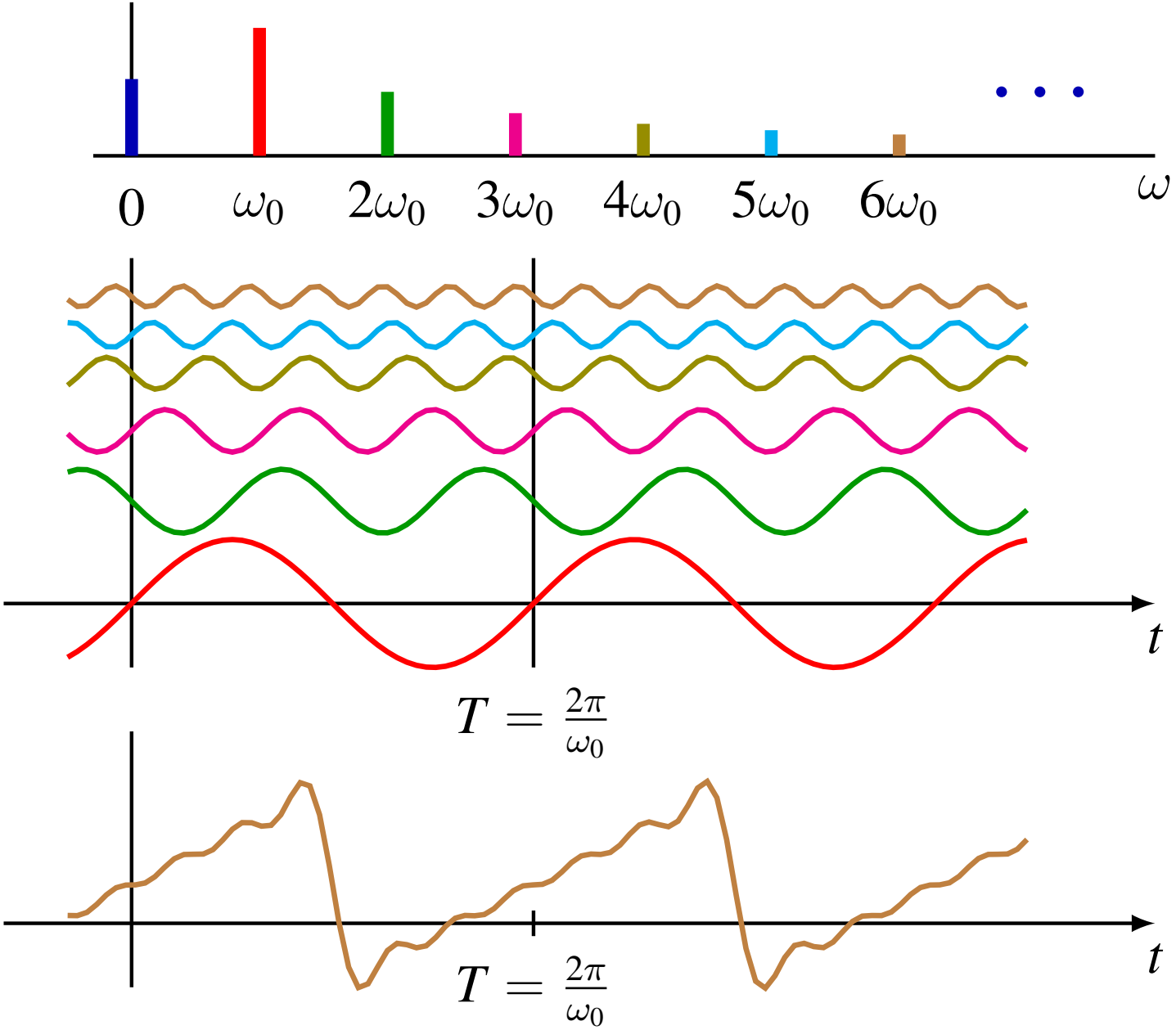
(https://gallica.bnf.fr)

Jean-Baptiste Joseph Fourier (from Wikipedia)

Harmonics


0
ω_0
$2\omega_0$
$3\omega_0$
$4\omega_0$
$5\omega_0$
$6\omega_0$
ω
0 1 2 3 4 5 6
$|k|$: harmonic #
DC
fundamental / first harmonic
second harmonic
third harmonic
fourth harmonic
fifth harmonic
sixth harmonic

Harmonics


0
ω_0
$2\omega_0$
$3\omega_0$
$4\omega_0$
$5\omega_0$
$6\omega_0$
ω
t
$T = \frac{2\pi}{\omega_0}$
t
$T = \frac{2\pi}{\omega_0}$

Orthonormality of Harmonics

Recall for $k \neq 0$,

$$\int_a^b e^{jk\omega_0 t} dt = \frac{e^{jk\omega_0 b} - e^{jk\omega_0 a}}{jk\omega_0}$$

Since $e^{jk\omega_0 t}$ has period T, over any period

$$\frac{1}{T}\int_T e^{jk\omega_0 t} dt = \frac{1}{T}\int_{t_0}^{t_0+T} e^{jk\omega_0 t} dt = \delta[k]$$

Define inner product between two signals with period T by

$$\langle f, g \rangle = \frac{1}{T}\int_T f(t)\overline{g(t)} dt$$

$\{e^{jk\omega_0 t} : k \in \mathbb{Z}\}$ is orthonormal system of functions

$$\langle e^{jk\omega_0 t}, e^{jm\omega_0 t} \rangle = \frac{1}{T}\int_T e^{j(k-m)\omega_0 t} dt = \delta_{km} = \delta[k-m]$$

Orthogonality of Harmonics

For sines and cosines,

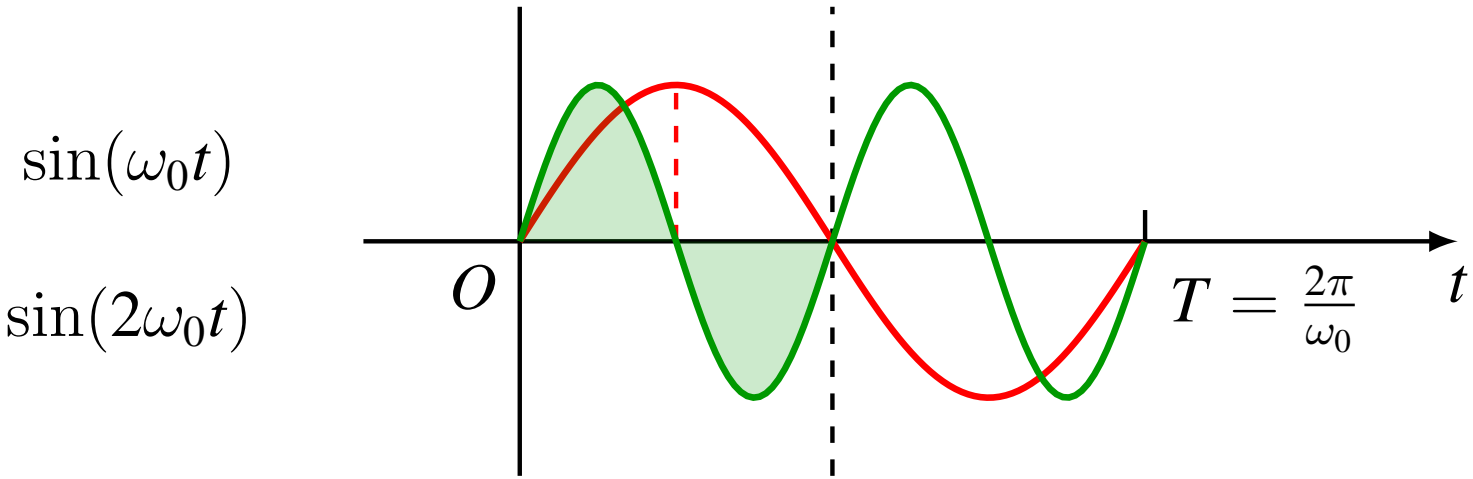
$$\langle \sin(k\omega_0 t), \sin(m\omega_0 t) \rangle = \frac{1}{2}\delta[k-m] - \frac{1}{2}\delta[k+m]$$

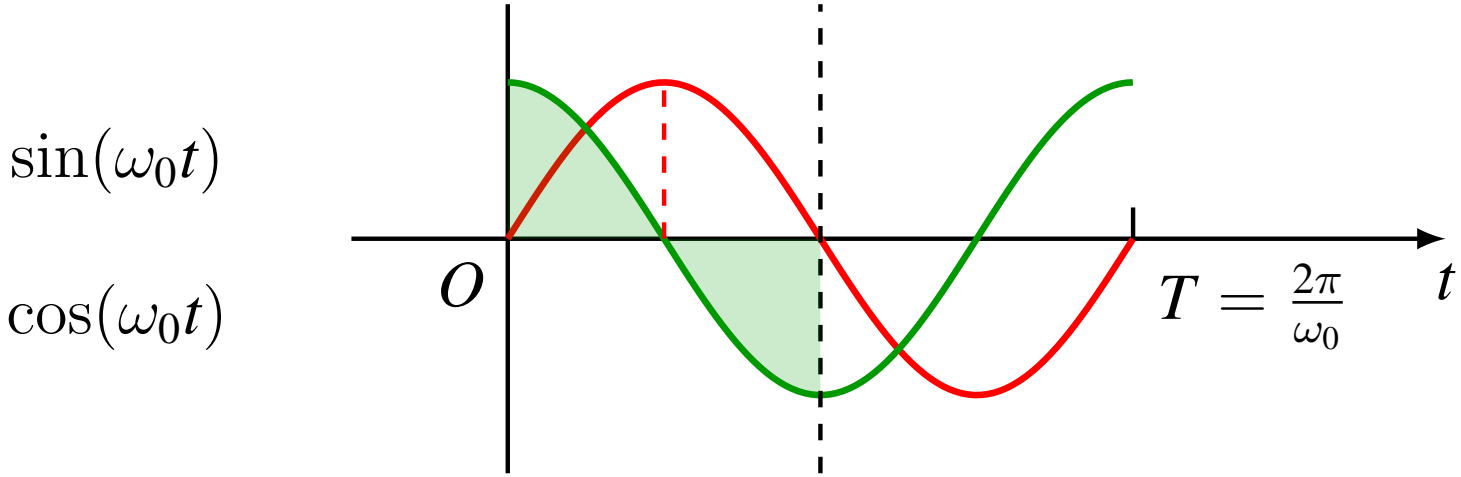
$$\langle \cos(k\omega_0 t), \cos(m\omega_0 t) \rangle = \frac{1}{2}\delta[k-m] + \frac{1}{2}\delta[k+m]$$

$$\langle \sin(k\omega_0 t), \cos(m\omega_0 t) \rangle = 0$$

Proof.
Use Euler's formula and orthonormality of $\{e^{jk\omega_0 t} : k \in \mathbb{Z}\}$

Orthogonality of Harmonics





Fourier Coefficients

Suppose x has period T and Fourier series representation

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}, \qquad \omega_0 = \frac{2\pi}{T}$$

Find Fourier coefficients using orthonormality of $\{e^{jk\omega_0 t}\}$

$$\begin{aligned}
\langle x, e^{jm\omega_0 t}\rangle &= \langle \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}, e^{jm\omega_0 t}\rangle \\
&= \sum_{k=-\infty}^{\infty} c_k \langle e^{jk\omega_0 t}, e^{jm\omega_0 t}\rangle \\
&= \sum_{k=-\infty}^{\infty} c_k \delta[m-k] = c_m
\end{aligned}$$

Complex Fourier Series

Synthesis equation

$$\boxed{x(t) = \sum_{k=-\infty}^{\infty} \hat{x}[k] e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} \hat{x}[k] e^{jk\frac{2\pi}{T}t}}$$

N-th partial sum

$$S_N(x)(t) = \sum_{k=-N}^{N} \hat{x}[k] e^{jk\omega_0 t}$$

Analysis equation

$$\boxed{\hat{x}[k] = \langle x, e^{jk\omega_0 t} \rangle = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk\frac{2\pi}{T}t} dt}$$

Trigonometric Fourier Series

For sines and cosines,

$$\langle \sin(k\omega_0 t), \sin(m\omega_0 t) \rangle = \frac{1}{2}\delta[k-m] - \frac{1}{2}\delta[k+m]$$
$$\langle \cos(k\omega_0 t), \cos(m\omega_0 t) \rangle = \frac{1}{2}\delta[k-m] + \frac{1}{2}\delta[k+m]$$
$$\langle \sin(k\omega_0 t), \cos(m\omega_0 t) \rangle = 0$$

Synthesis equation

$$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)]$$

Analysis equation

$$a_k = \frac{2-\delta[k]}{T} \int_T x(t)\cos(k\omega_0 t)dt, \quad b_k = \frac{2}{T}\int_T x(t)\sin(k\omega_0 t)dt$$

Equivalence of Two Forms

Complex form

$$x(t) = \sum_{k=-\infty}^{\infty} \hat{x}[k] e^{jk\omega_0 t}$$

Trigonometric form

$$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)]$$

Conversion of coefficients (by Euler's formula)

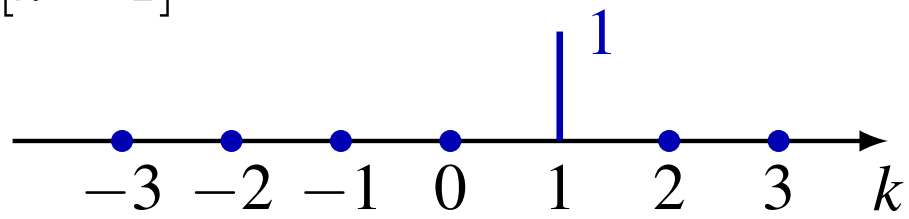
$$\begin{cases} a_0 = \hat{x}[0] & \\ a_k = \hat{x}[k] + \hat{x}[-k], & k \geq 1 \\ b_k = j(\hat{x}[k] - \hat{x}[-k]), & k \geq 1 \end{cases} \qquad \begin{cases} \hat{x}[0] = a_0 & \\ \hat{x}[k] = \frac{1}{2}(a_k - jb_k), & k \geq 1 \\ \hat{x}[k] = \frac{1}{2}(a_k + jb_{-k}), & k \leq -1 \end{cases}$$

Negative frequencies in complex form introduced for mathematical convenience, no physical significance

Examples

Example. $x(t) = e^{j\omega_0 t}$, $\hat{x}[k] = \delta[k-1]$

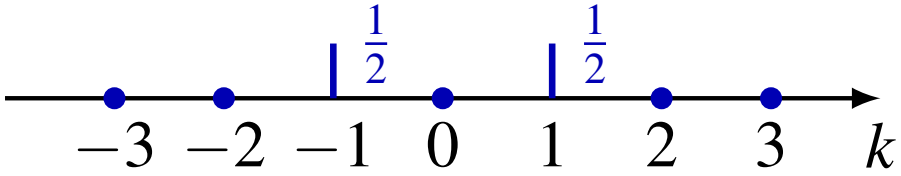
Spectrum $\hat{x}[k]$



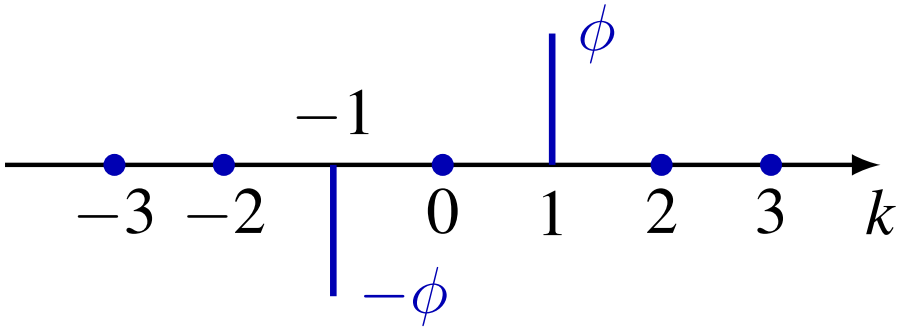
Example. $x(t) = \cos(\omega_0 t + \phi) = \frac{e^{j\phi}}{2} e^{j\omega_0 t} + \frac{e^{-j\phi}}{2} e^{-j\omega_0 t}$,

$$\hat{x}[1] = \frac{1}{2} e^{j\phi}, \quad , \hat{x}[-1] = \frac{1}{2} e^{-j\phi}, \quad \hat{x}[k] = 0, k \neq \pm 1$$

Magnitude spectrum $|\hat{x}[k]|$



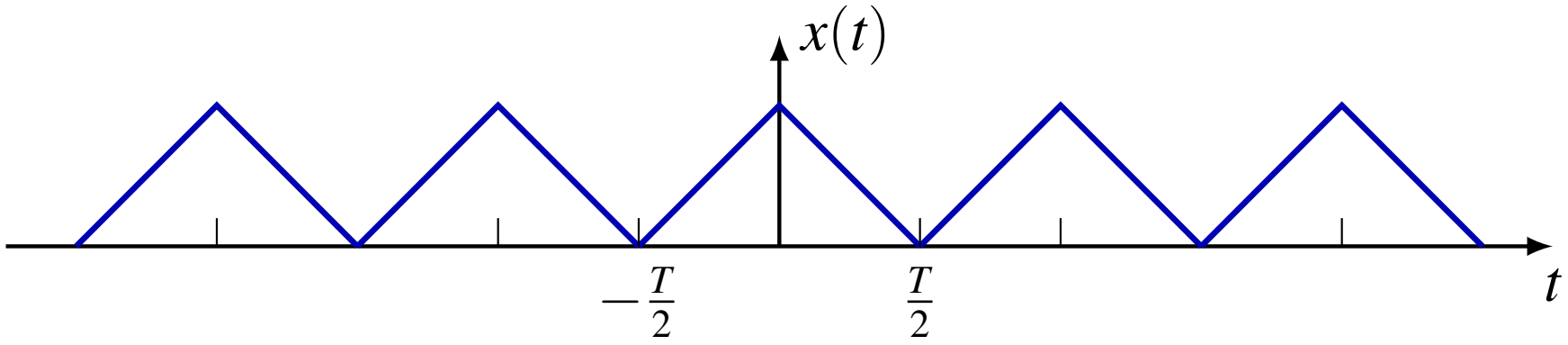
Phase spectrum $\arg \hat{x}[k]$



Example: Triangle Wave

In one period,

$$x(t) = 1 - \frac{2|t|}{T}, \quad |t| \le \frac{T}{2}$$



Fourier coefficients

$$k = 0, \quad \hat{x}[0] = \frac{1}{T}\int_{-T/2}^{T/2}\left(1 - \frac{2|t|}{T}\right)dt = \frac{1}{2}$$

$$k \neq 0, k \text{ odd}, \quad \hat{x}[k] = \frac{1}{T}\int_{-T/2}^{T/2}\left(1 - \frac{2|t|}{T}\right)e^{-jk\omega_0 t}dt = \frac{2}{\pi^2 k^2}$$

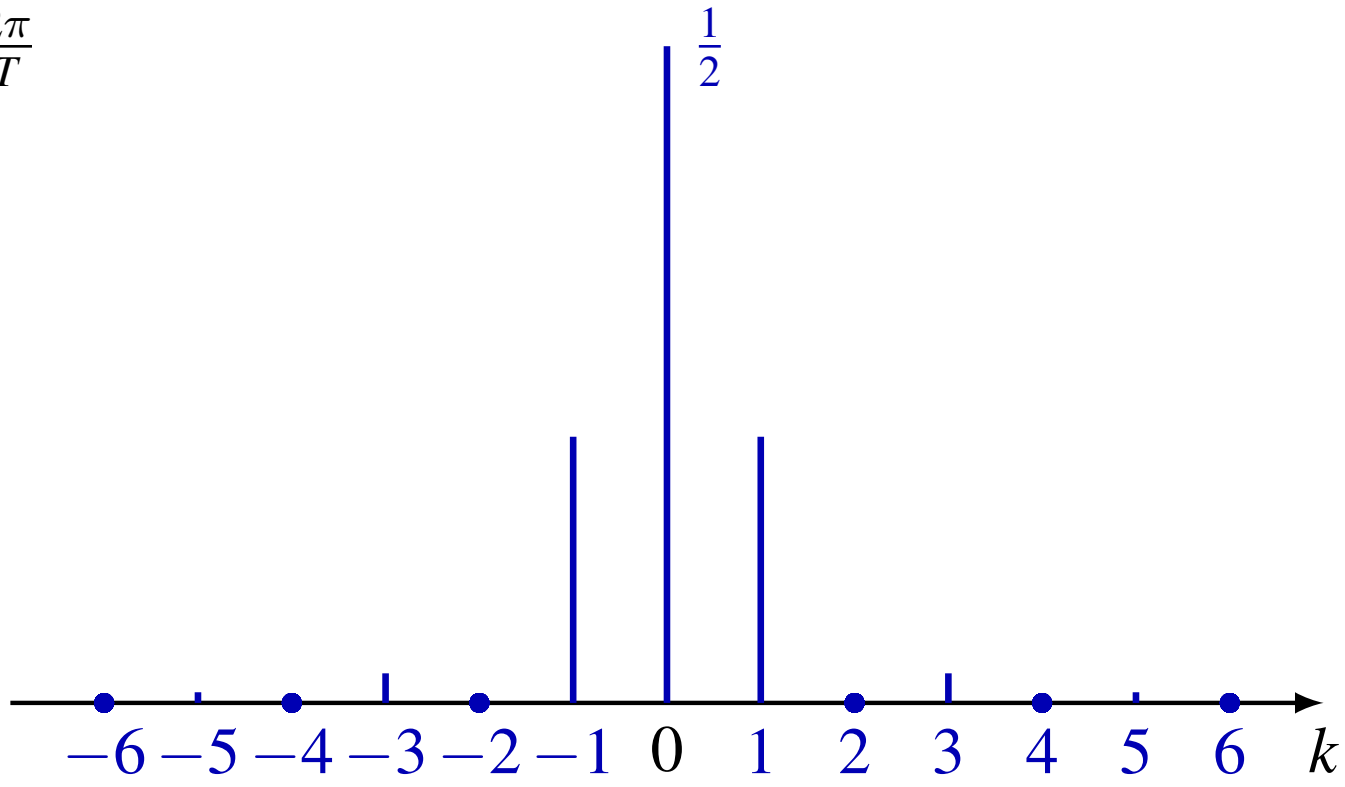
$$k \neq 0, k \text{ even}, \quad \hat{x}[k] = 0$$

Example: Triangle Wave

$$\hat{x}[k] = \begin{cases} \frac{1}{2}, & k = 0 \\ \frac{2}{\pi^2 k^2}, & k \neq 0 \text{ odd} \\ 0, & k \neq 0 \text{ even} \end{cases}$$

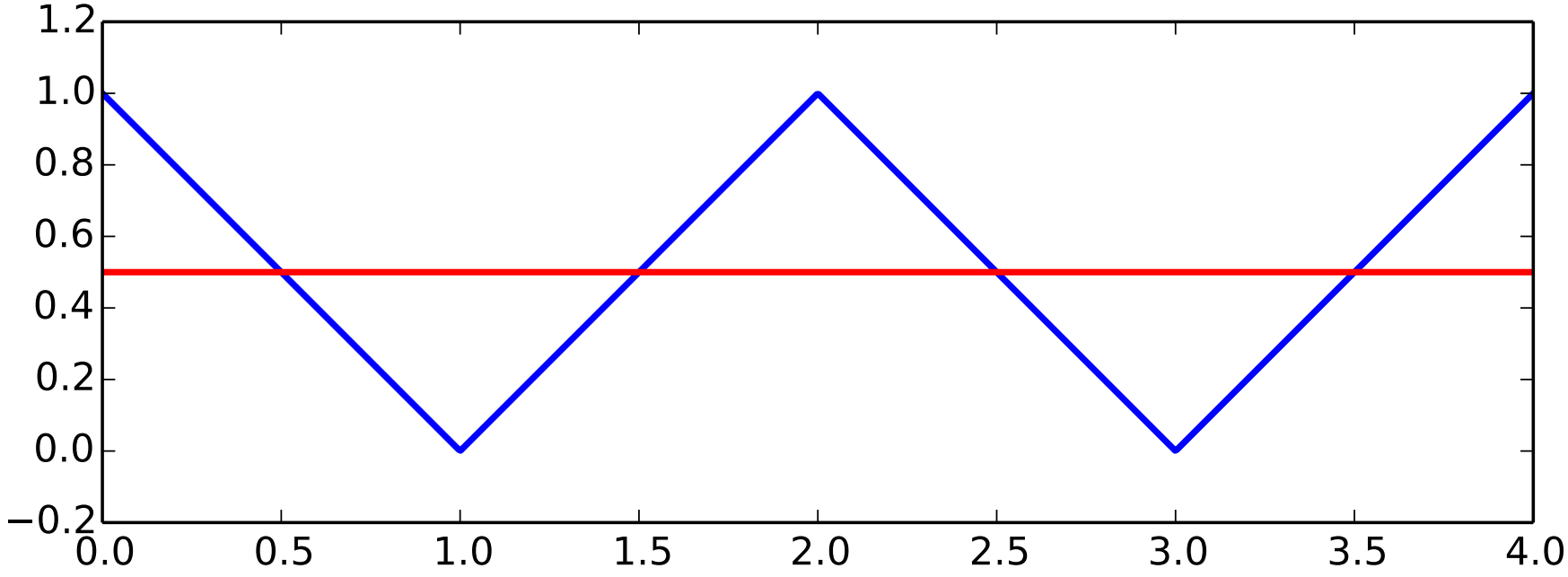
Spectrum for fixed T, frequency spacing $\Delta\omega = \frac{2\pi}{T}$

$\omega_k = k\frac{2\pi}{T}$



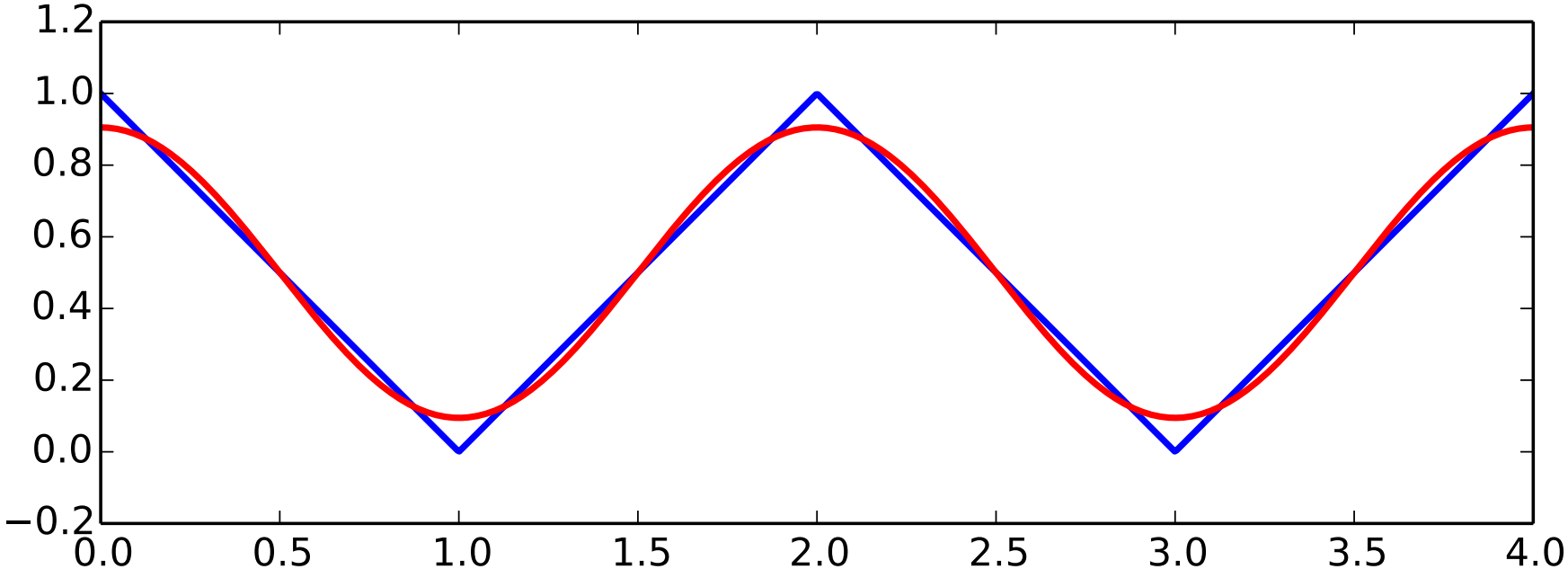
Example: Triangle Wave

$$S_0(x)(t) = \sum_{k=-0}^{0} \hat{x}[k] e^{jk\omega_0 t}$$



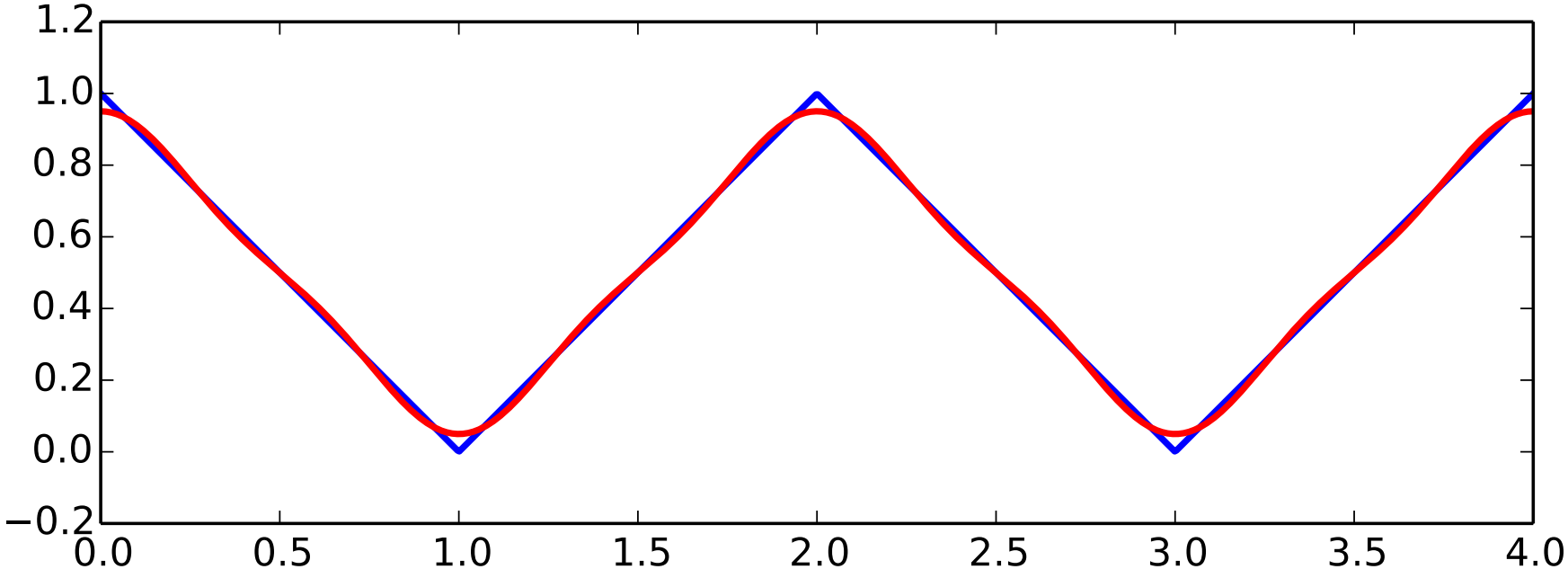
Example: Triangle Wave

$$S_1(x)(t) = \sum_{k=-1}^{1} \hat{x}[k] e^{jk\omega_0 t}$$



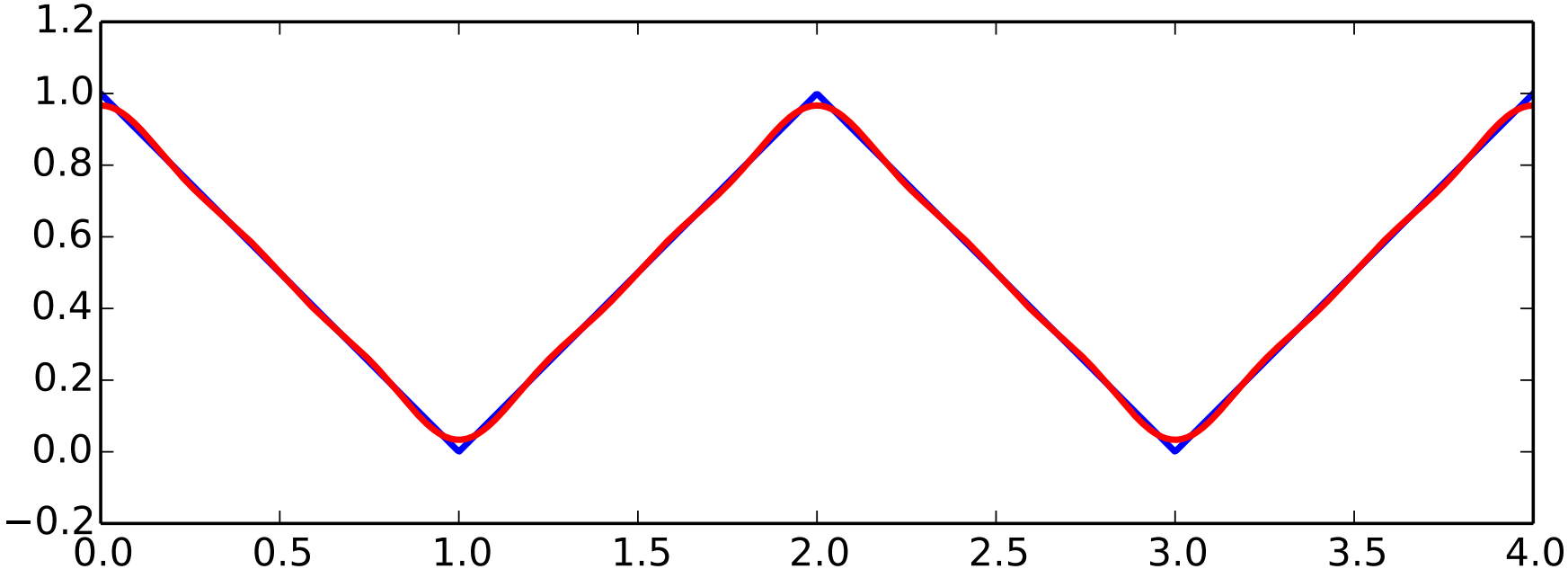
Example: Triangle Wave

$$S_3(x)(t) = \sum_{k=-3}^{3} \hat{x}[k] e^{jk\omega_0 t}$$



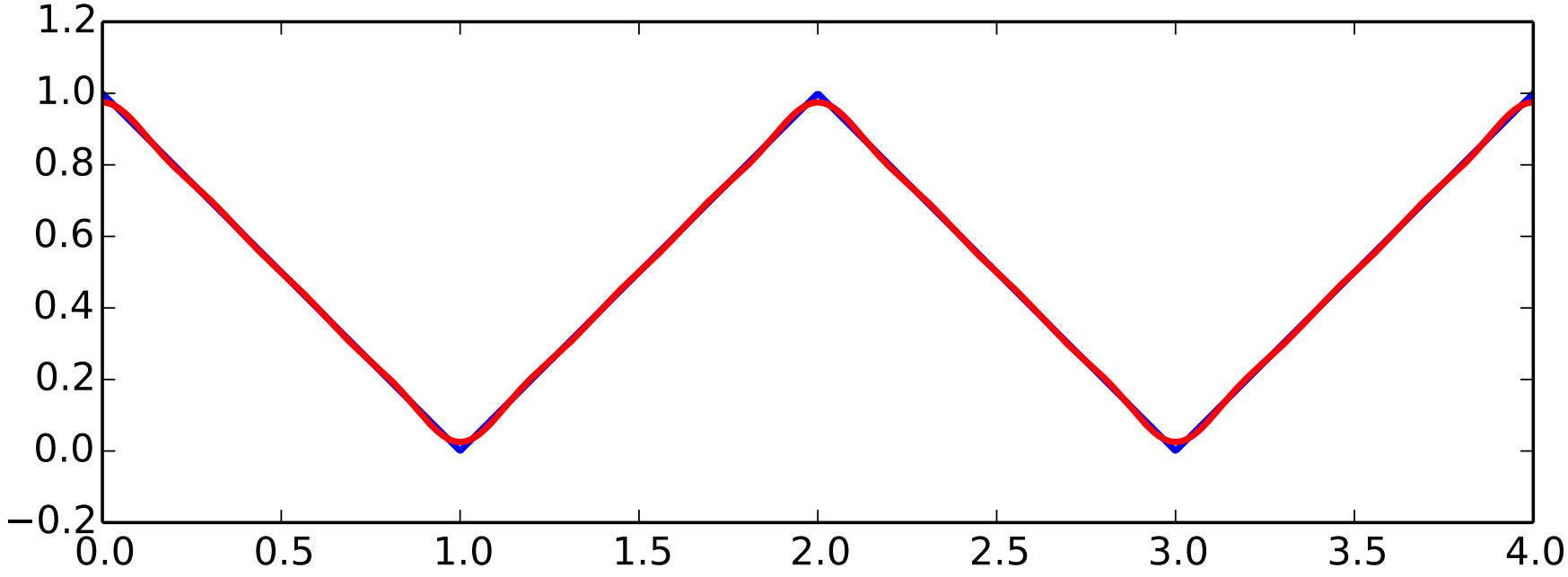
Example: Triangle Wave

$$S_5(x)(t) = \sum_{k=-5}^{5} \hat{x}[k] e^{jk\omega_0 t}$$



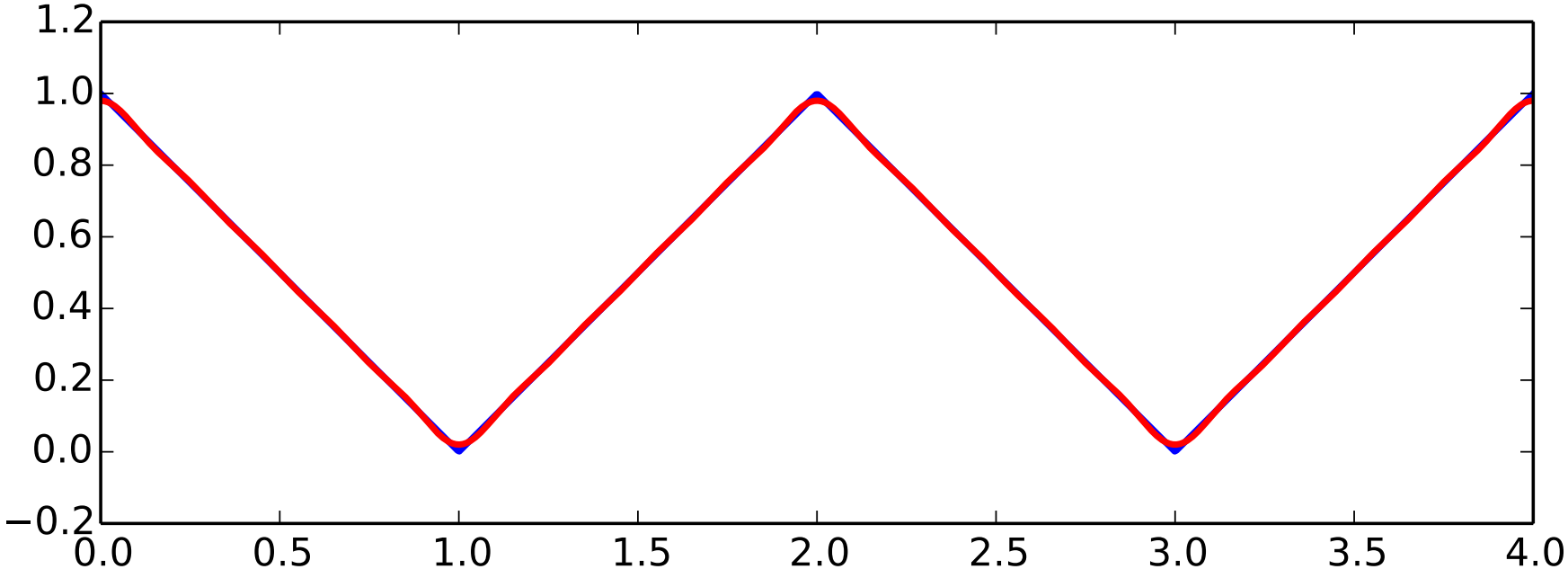
Example: Triangle Wave

$$S_7(x)(t) = \sum_{k=-7}^{7} \hat{x}[k] e^{jk\omega_0 t}$$



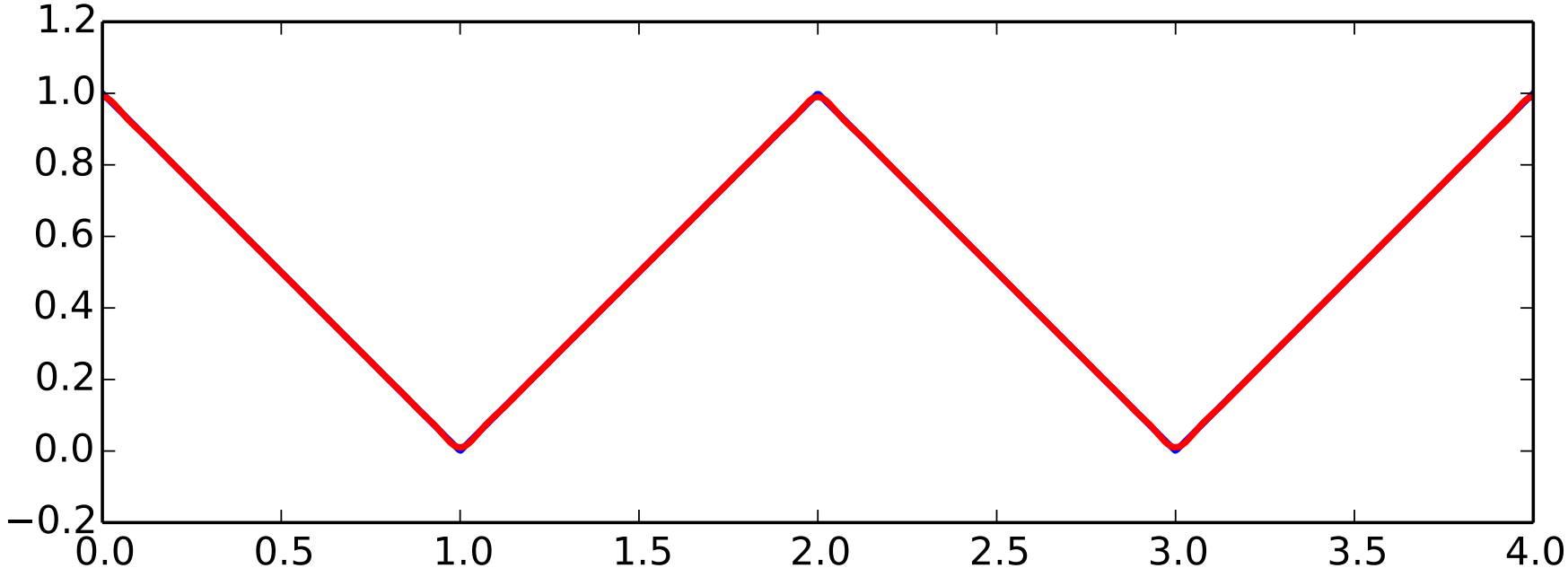
Example: Triangle Wave

$$S_9(x)(t) = \sum_{k=-9}^{9} \hat{x}[k] e^{jk\omega_0 t}$$



Example: Triangle Wave

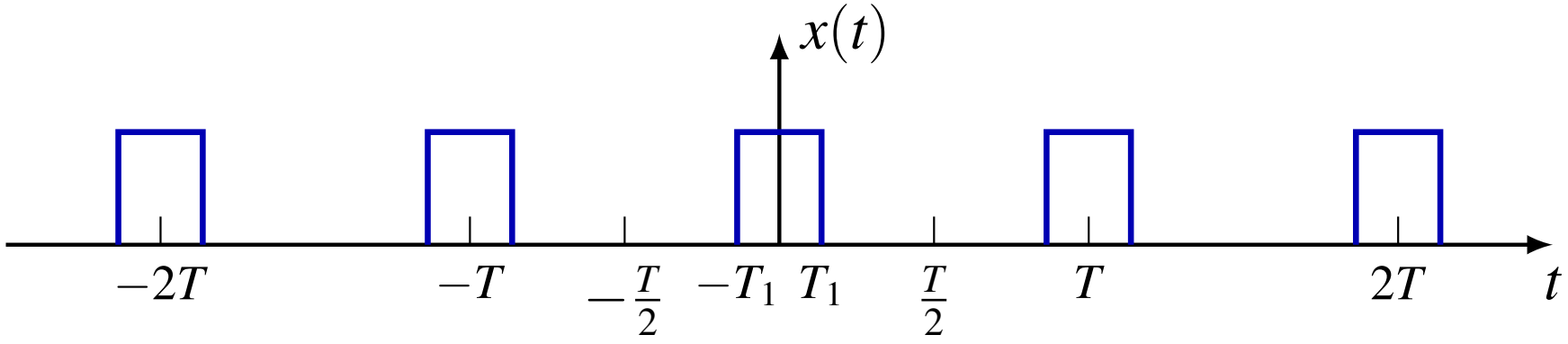
$$S_{19}(x)(t) = \sum_{k=-19}^{19} \hat{x}[k] e^{jk\omega_0 t}$$



Example: Periodic Square Wave

In one period,

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$



Fourier coefficients

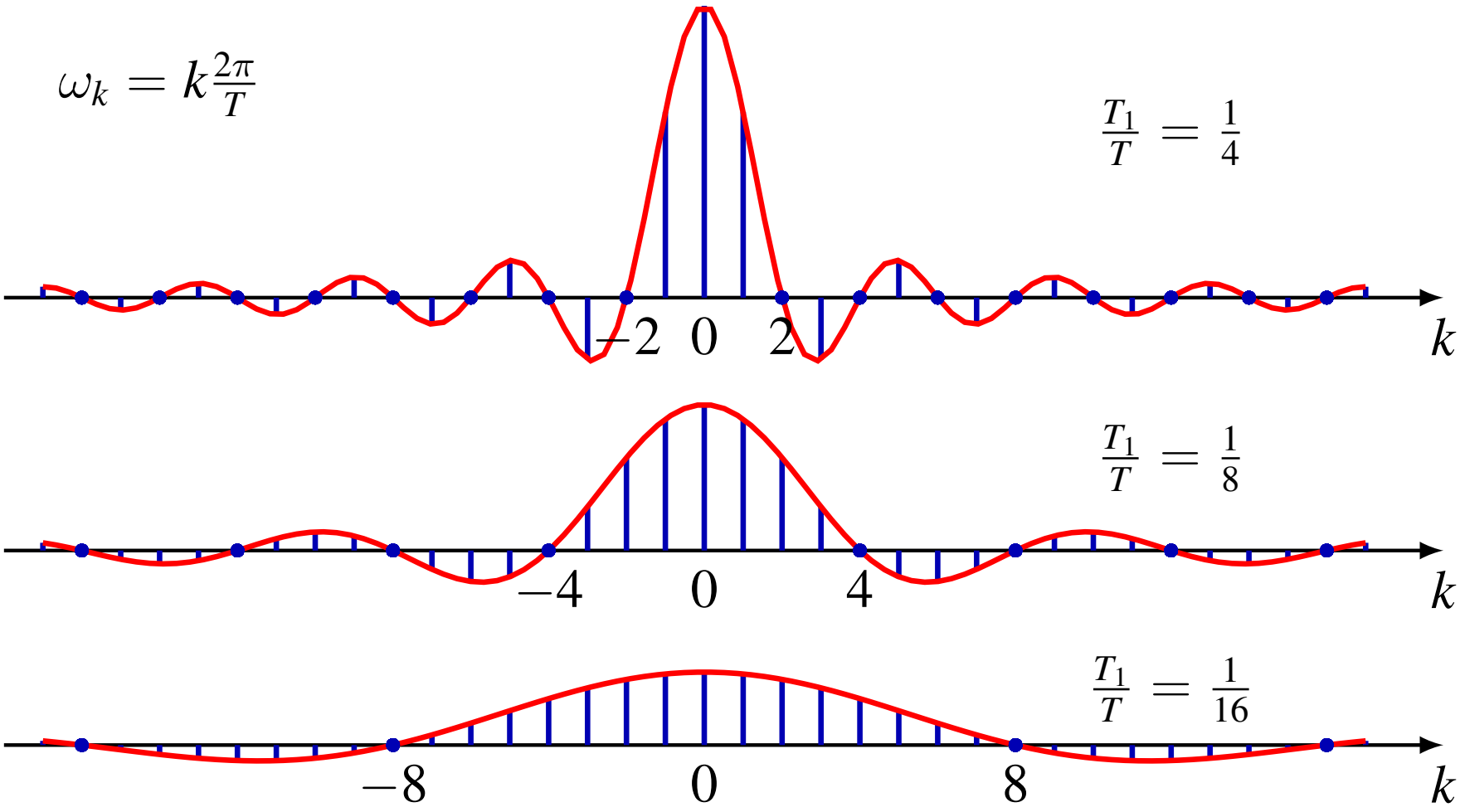
$$k = 0, \quad \hat{x}[0] = \frac{1}{T}\int_{-T_1}^{T_1} 1 dt = \frac{2T_1}{T}$$

$$k \neq 0, \quad \hat{x}[k] = \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(k\omega_0 T_1)}{k\pi}$$

Example: Periodic Square Wave

$$\hat{x}[k] = \frac{\sin\left(k\frac{2\pi T_1}{T}\right)}{k\pi} = \frac{2\sin(\omega_k T_1)}{\omega_k T}$$

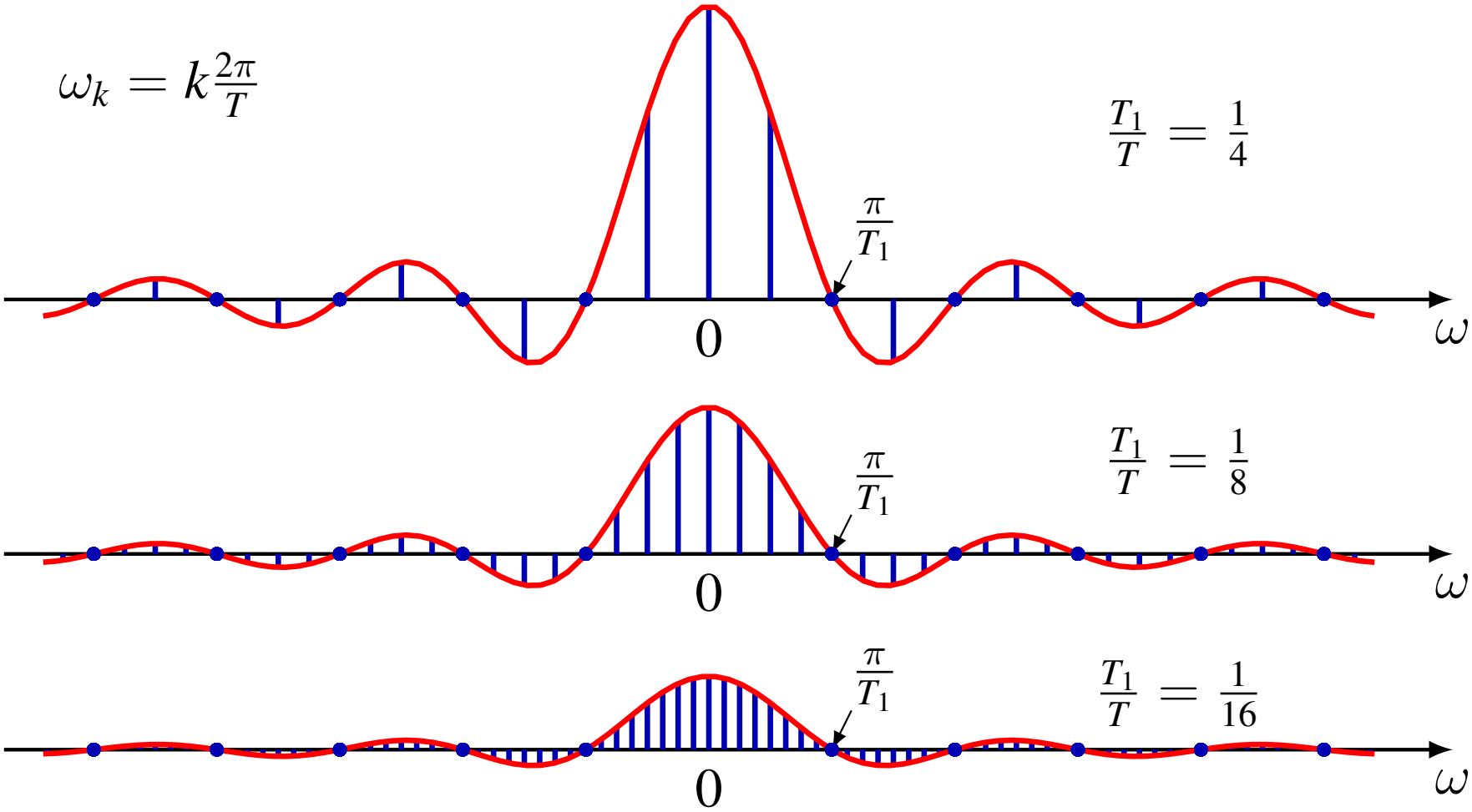
Spectra for fixed T and different T_1



Example: Periodic Square Wave

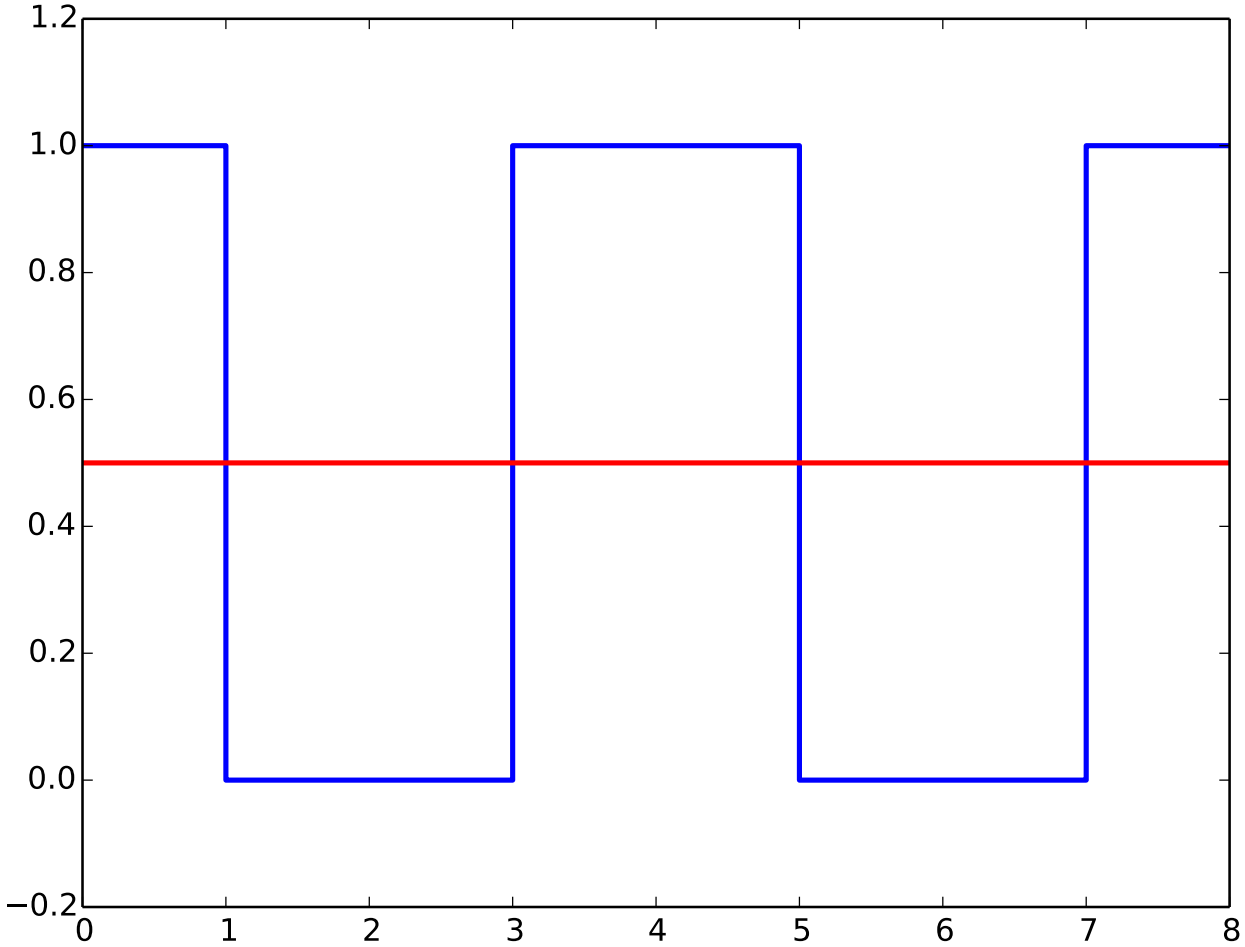
$$\hat{x}[k] = \frac{\sin\left(k\frac{2\pi T_1}{T}\right)}{k\pi} = \frac{2\sin(\omega_k T_1)}{\omega_k T}$$

Spectra for fixed T_1 and different T



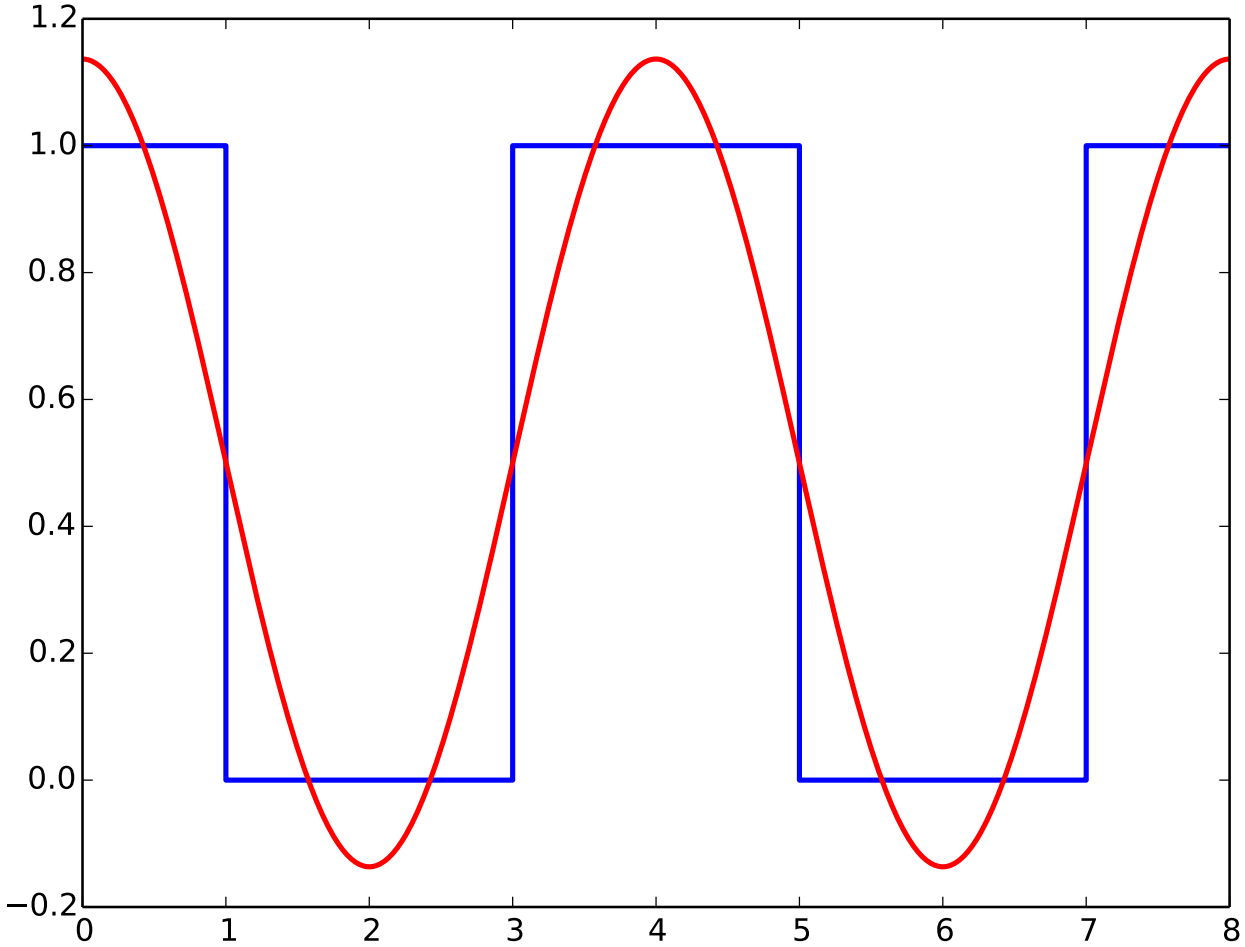
Example: Periodic Square Wave

$$S_0(x)(t) = \sum_{k=-0}^{0} \hat{x}[k] e^{jk\omega_0 t}$$



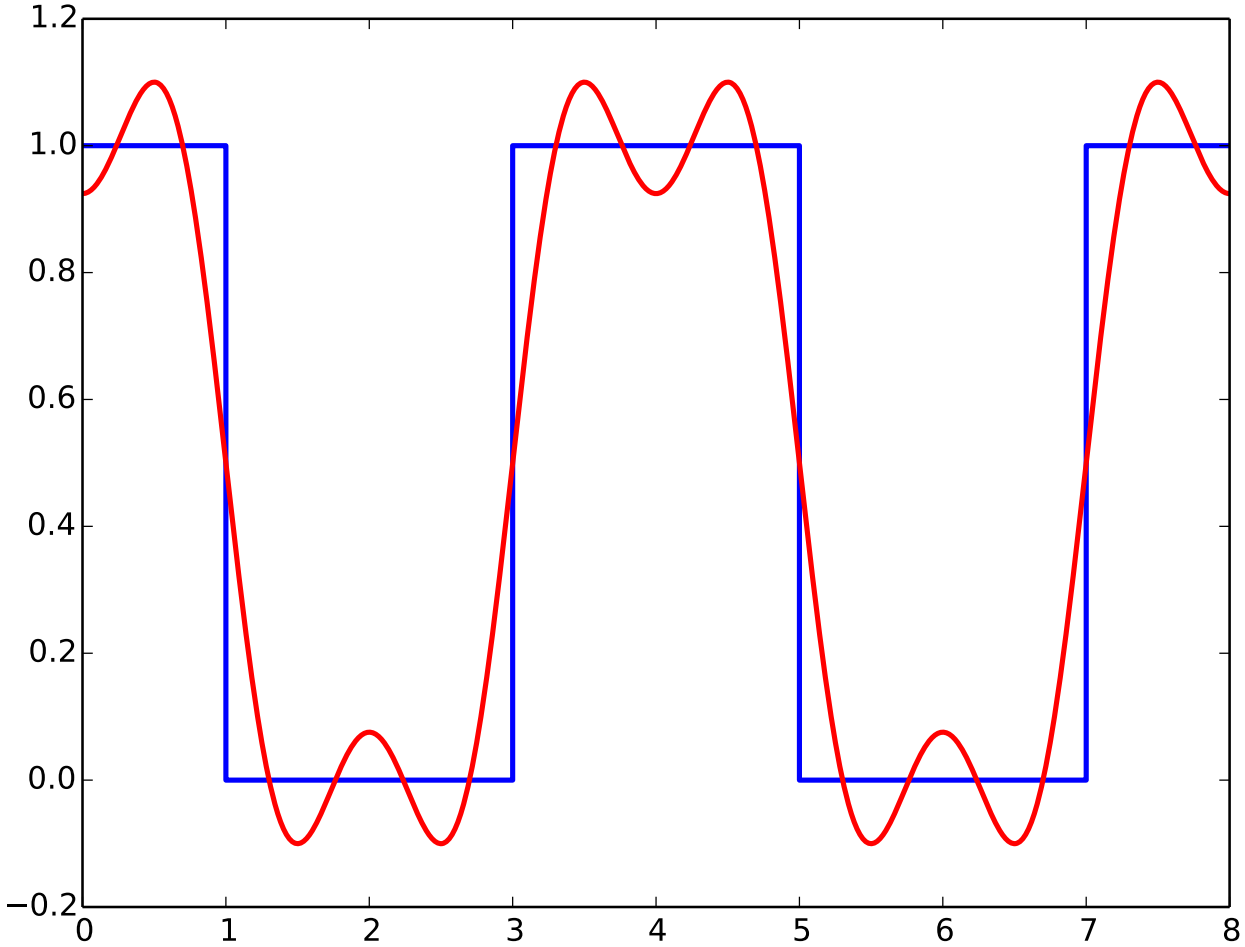
Example: Periodic Square Wave

$$S_1(x)(t) = \sum_{k=-1}^{1} \hat{x}[k] e^{jk\omega_0 t}$$



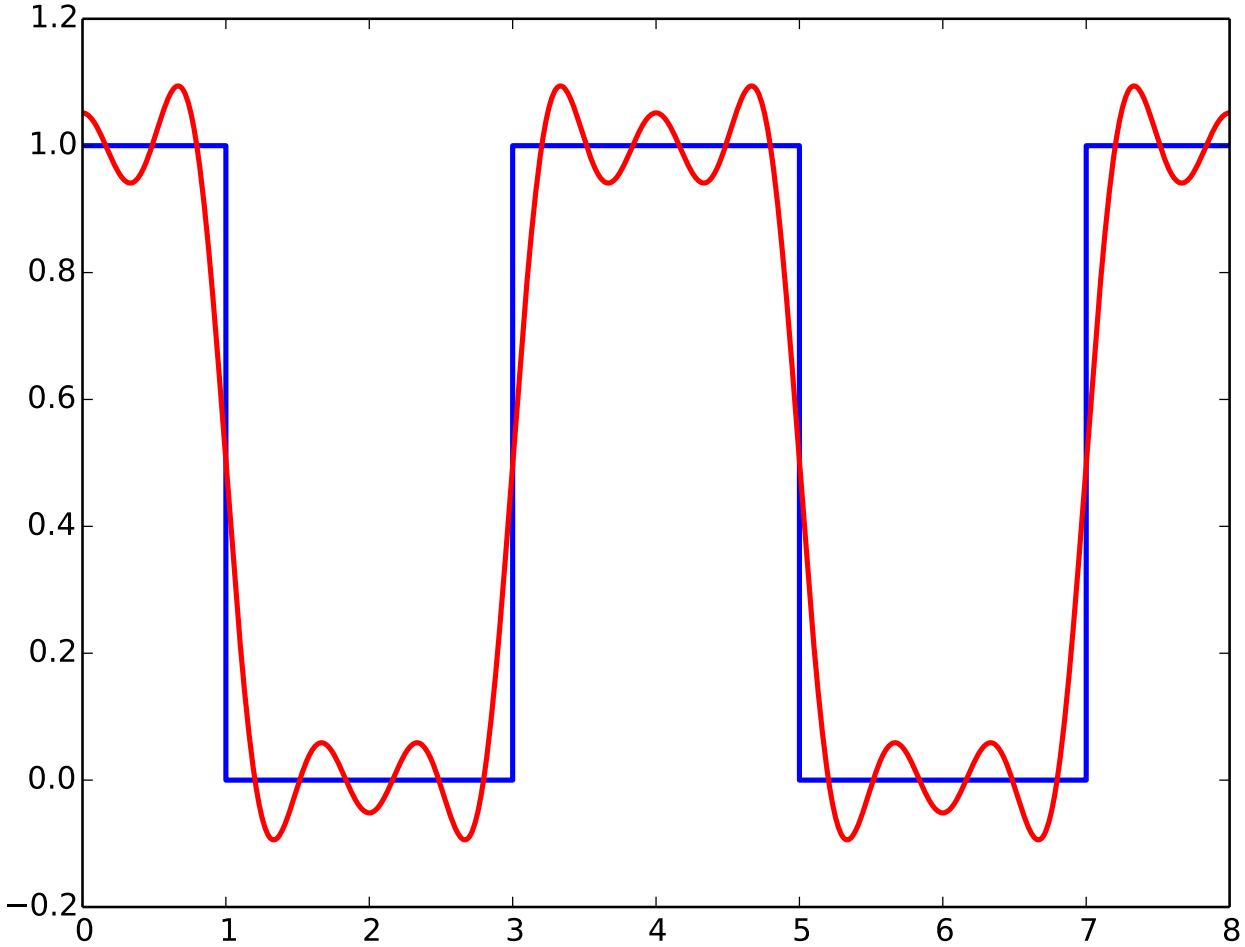
Example: Periodic Square Wave

$$S_3(x)(t) = \sum_{k=-3}^{3} \hat{x}[k] e^{jk\omega_0 t}$$



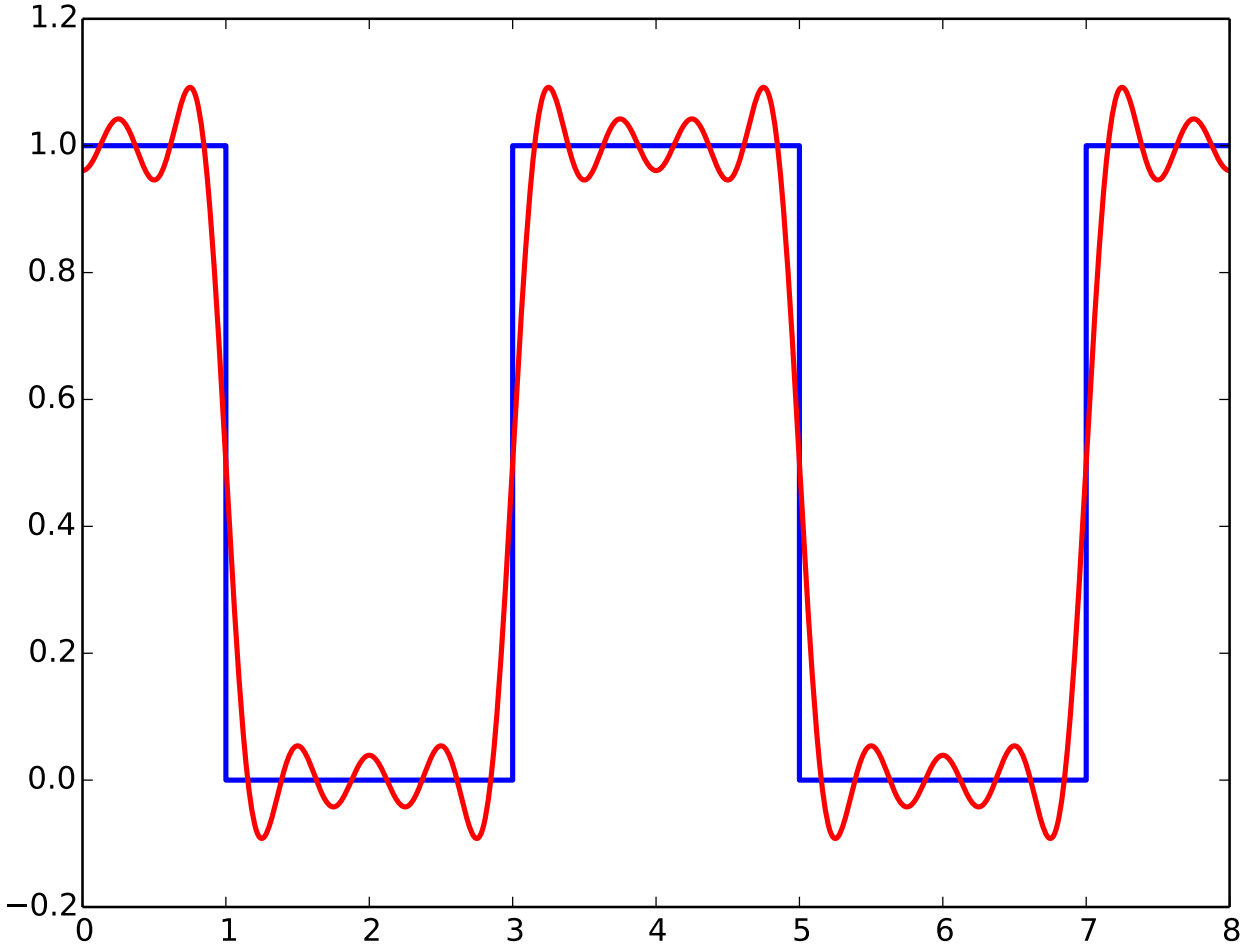
Example: Periodic Square Wave

$$S_5(x)(t) = \sum_{k=-5}^{5} \hat{x}[k] e^{jk\omega_0 t}$$



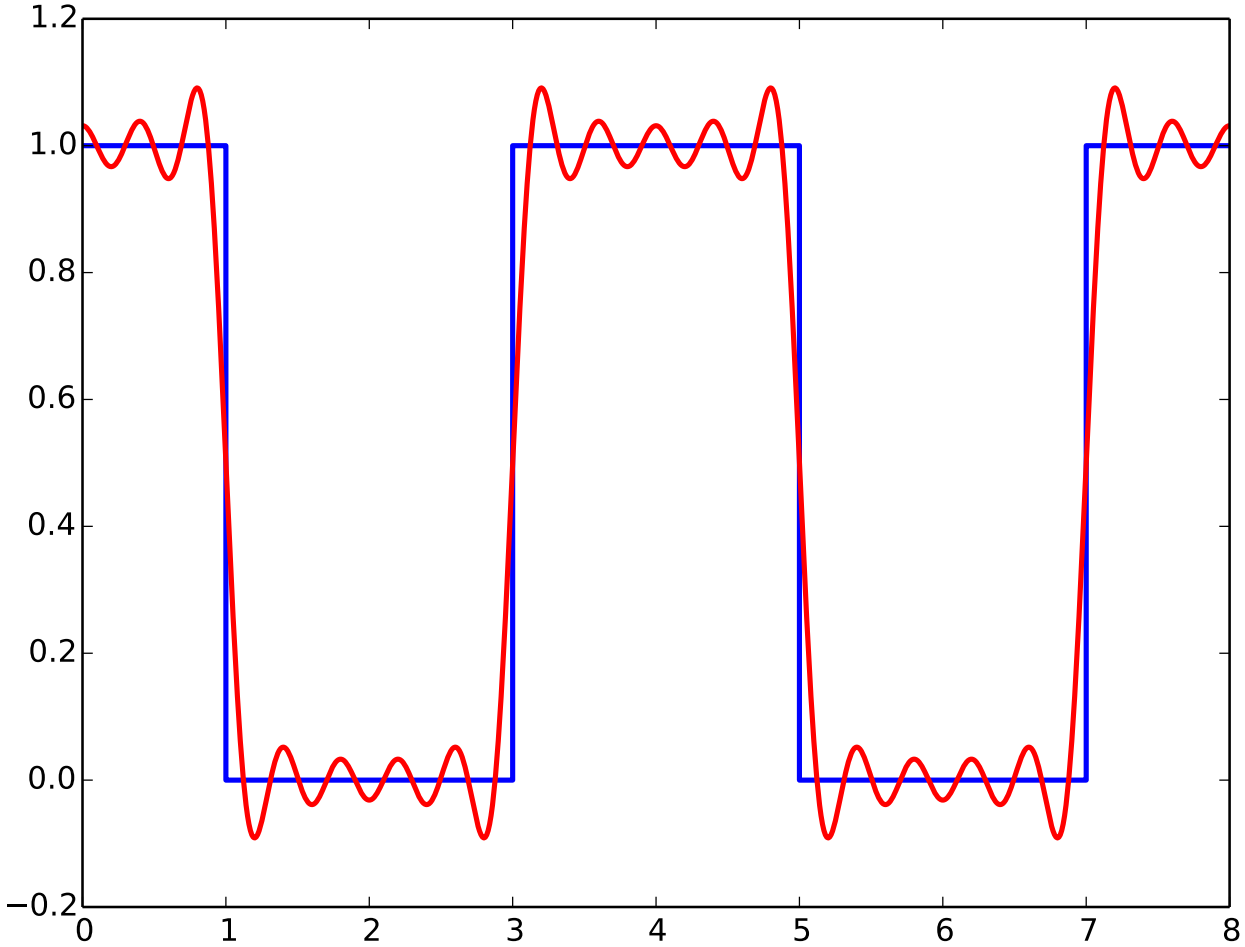
Example: Periodic Square Wave

$$S_7(x)(t) = \sum_{k=-7}^{7} \hat{x}[k] e^{jk\omega_0 t}$$



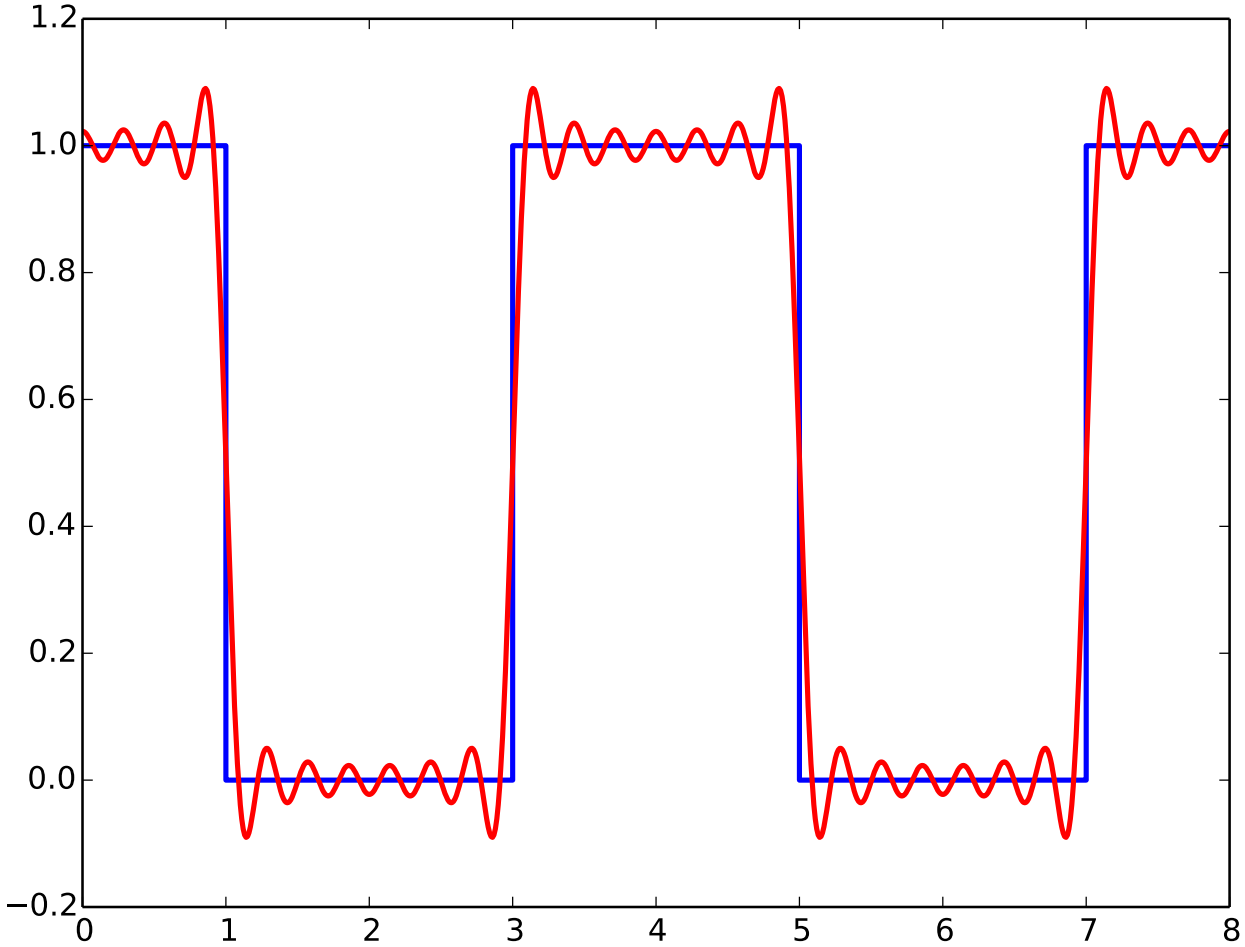
Example: Periodic Square Wave

$$S_9(x)(t) = \sum_{k=-9}^{9} \hat{x}[k] e^{jk\omega_0 t}$$



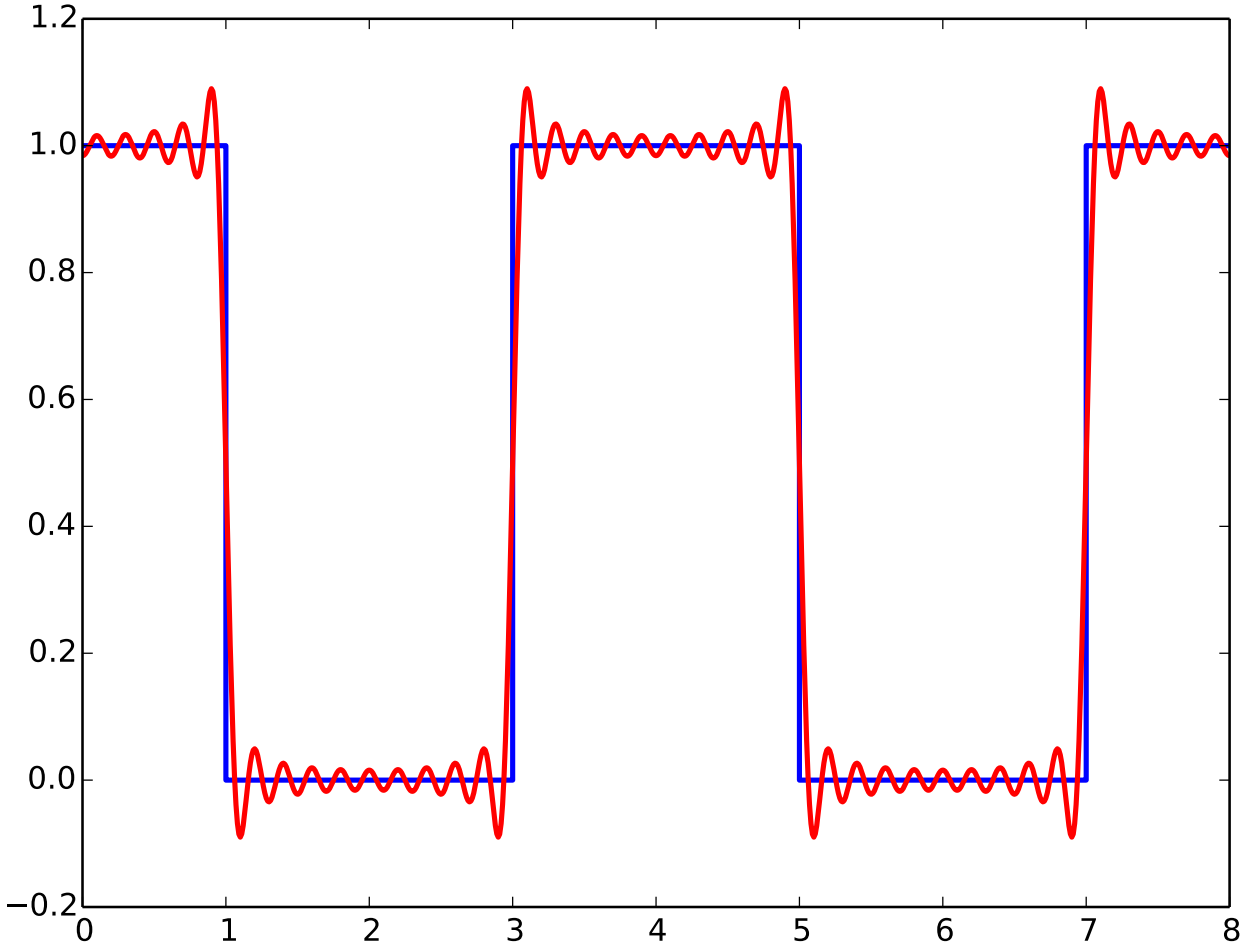
Example: Periodic Square Wave

$$S_{13}(x)(t) = \sum_{k=-13}^{13} \hat{x}[k] e^{jk\omega_0 t}$$



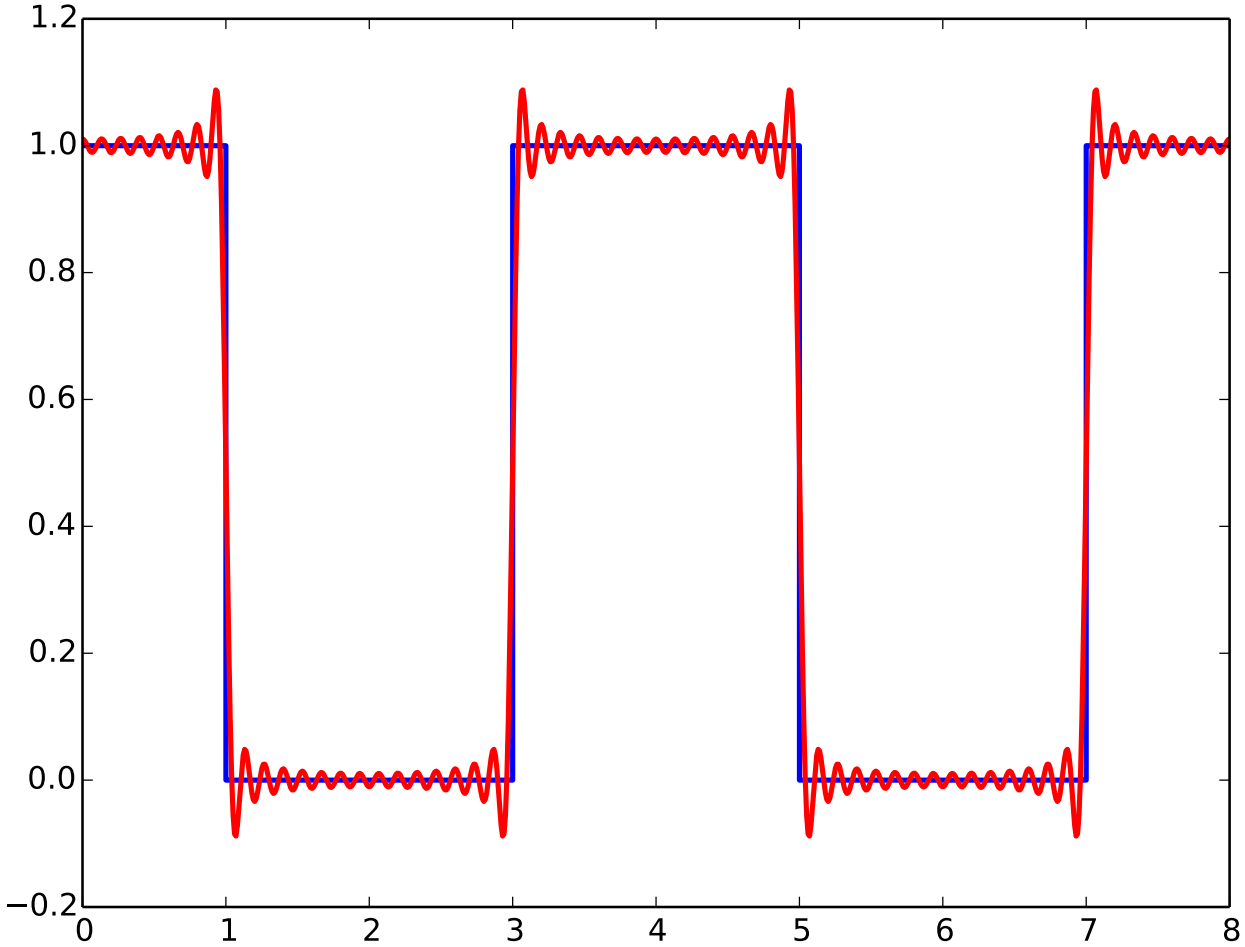
Example: Periodic Square Wave

$$S_{19}(x)(t) = \sum_{k=-19}^{19} \hat{x}[k] e^{jk\omega_0 t}$$



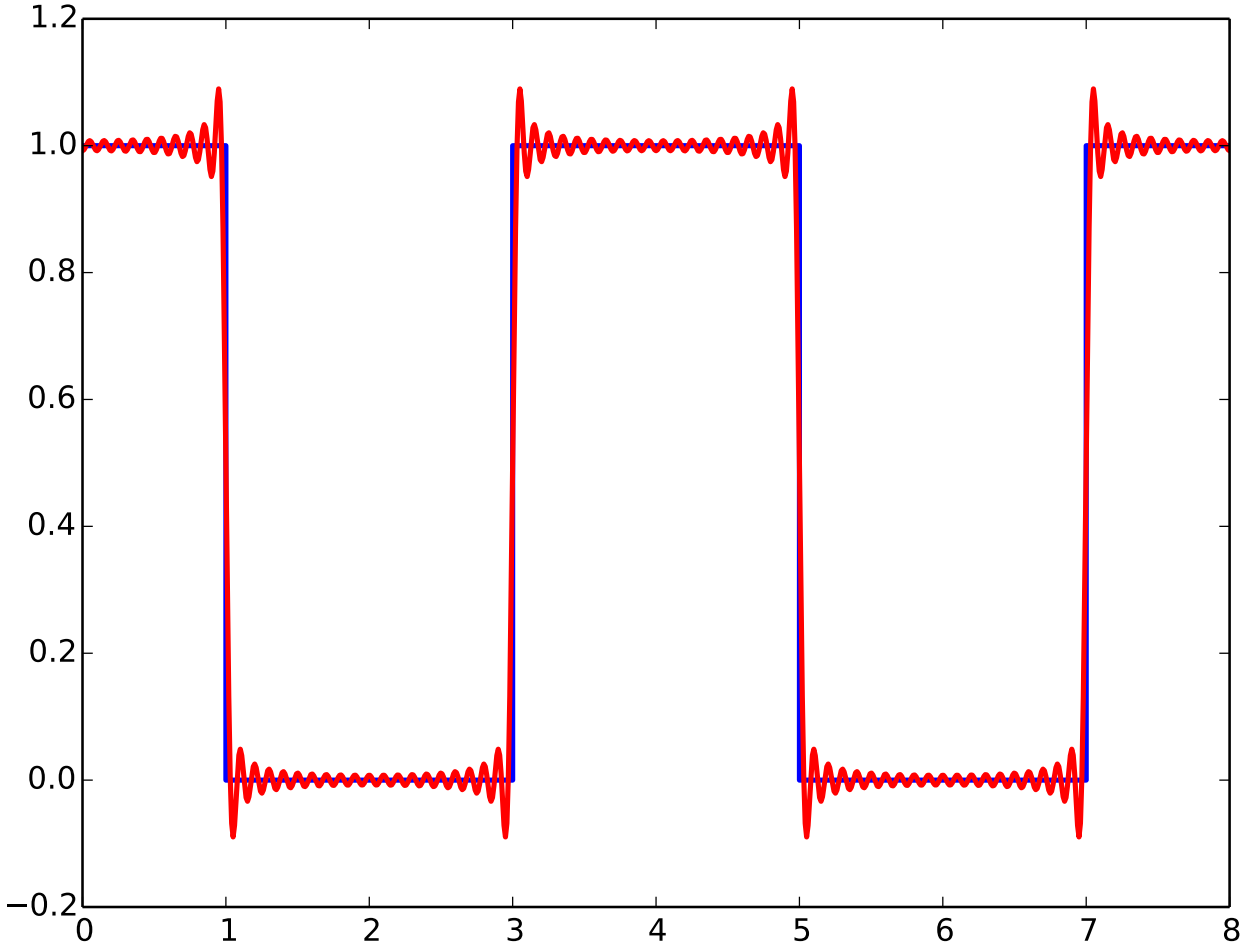
Example: Periodic Square Wave

$$S_{29}(x)(t) = \sum_{k=-29}^{29} \hat{x}[k] e^{jk\omega_0 t}$$



Example: Periodic Square Wave

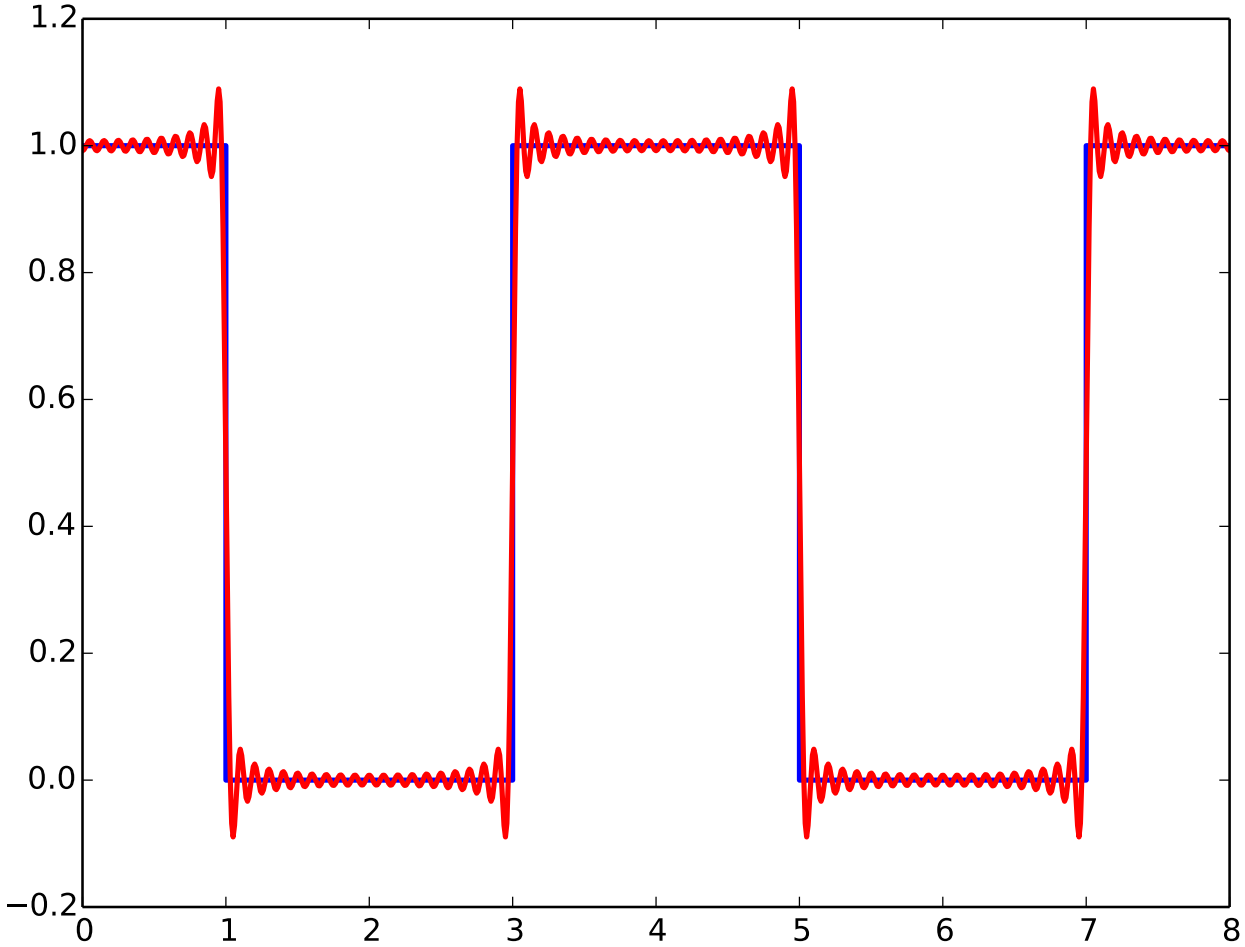
$$S_{39}(x)(t) = \sum_{k=-39}^{39} \hat{x}[k] e^{jk\omega_0 t}$$



Gibbs Phenomenon

Partial sums of Fourier series "ring" at jump discontinuity

- overshoot gets closer and closer to discontinuity
- overshoot approaches $\approx 9\%$ of jump size



Contents

1. Eigenvalues and Eigenfunctions

2. CT Fourier Series

3. Properties of CT Fourier Series

Properties of CT Fourier Series

Fourier series establishes correspondence between periodic functions and doubly infinite sequences

$$x \xleftrightarrow{\mathcal{FS}} \hat{x} \qquad \text{or} \qquad x(t) \xleftrightarrow{\mathcal{FS}} \hat{x}[k]$$

Linearity

If x, y have **same** period T, so does their linear combination $ax + by$, and

$$\widehat{ax + by} = a\hat{x} + b\hat{y}$$

Proof.

$$\begin{aligned}(\widehat{ax + by})[k] &= \langle ax + by, e^{jk\omega_0 t}\rangle \\ &= a\langle x, e^{jk\omega_0 t}\rangle + b\langle y, e^{jk\omega_0 t}\rangle \\ &= a\hat{x}[k] + b\hat{y}[k]\end{aligned}$$

Properties of CT Fourier Series

Time shifting

If x has period T and $\omega_0 = \frac{2\pi}{T}$,

$$\widehat{\tau_{t_0} x} = E_{-\omega_0 t_0} \hat{x} \qquad \text{or} \qquad x(t - t_0) \xleftrightarrow{\mathcal{FS}} e^{-jk\omega_0 t_0} \hat{x}[k]$$

where $(E_a \hat{x})[k] = e^{jak} \hat{x}[k]$

Time shift $\iff$ linear phase change in frequency

Proof.

$$\begin{aligned}
(\widehat{\tau_{t_0} x})[k] &= \langle \tau_{t_0} x, e^{jk\omega_0 t} \rangle = \langle x, \tau_{-t_0} e^{jk\omega_0 t} \rangle \\
&= \langle x, e^{jk\omega_0 (t+t_0)} \rangle = \langle x, e^{jk\omega_0 t_0} e^{jk\omega_0 t} \rangle \\
&= e^{-jk\omega_0 t_0} \langle x, e^{jk\omega_0 t} \rangle = e^{-jk\omega_0 t_0} \hat{x}[k] = (E_{-\omega_0 t_0} \hat{x})[k]
\end{aligned}$$

Example. $\cos(t) = \frac{1}{2} e^{jt} + \frac{1}{2} e^{-jt}$, $\sin(t) = \cos(t - \frac{\pi}{2}) = \frac{-j}{2} e^{jt} + \frac{j}{2} e^{-jt}$